

**UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL
CENTRO DE BIOTECNOLOGIA
PROGRAMA DE PÓS-GRADUAÇÃO EM BIOLOGIA CELULAR E
MOLECULAR**

**CARACTERIZAÇÃO DA MICROBIOTA BACTERIANA E DO VIROMA
INTESTINAL DE LEITÕES COM E SEM DIARREIA PÓS-DESMAME**

Tese de doutorado

Autora: Karen Karine da Rosa Dias

Porto Alegre, janeiro de 2026

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Tese submetida ao Programa de Pós-Graduação em Biologia Celular e Molecular do Centro de Biotecnologia da Universidade Federal do Rio Grande do Sul como requisito parcial para obtenção do Título de Doutor em Ciências.

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Porto Alegre, janeiro de 2026

CIP - Catalogação na Publicação

da Rosa Dias, Karen Karine
CARACTERIZAÇÃO INTEGRADA DA MICROBIOTA BACTERIANA E DO
VIROMA INTESTINAL DE LEITÕES COM E SEM DIARREIA
POS-DESMAME / Karen Karine da Rosa Dias. -- 2026.
85 f.
Orientadora: Fabiana Quoos Mayer.

Tese (Doutorado) -- Universidade Federal do Rio Grande do Sul, Centro de
Biotecnologia do Estado do Rio Grande do Sul, Programa de Pós-Graduação em Biologia
Celular e Molecular, Porto Alegre, BR-RS, 2026.

1. Suinocultura. 2. Microbiota bacteriana. 3. Viroma. 4. Bioinformatica. 5. Gastrointestinal. I.
Quoos Mayer, Fabiana, orient. II. Título.

Elaborada pelo Sistema de Geração Automática de Ficha Catalográfica da UFRGS com os
dados fornecidos pelo(a) autor(a).

Este trabalho foi desenvolvido no Laboratório de Saúde Única, situado no Centro de Biotecnologia da Universidade Federal do Rio Grande do Sul com apoio financeiro do Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) e da Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES).

SUMÁRIO

LISTA DE ABREVIATURAS, SÍMBOLOS E UNIDADES.....	6
LISTA DE FIGURAS	7
RESUMO	8
ABSTRACT	10
1. INTRODUÇÃO	12
2. REVISÃO BIBLIOGRÁFICA	14
2.1 SUINOCULTURA	14
2.1.1 Desmame precoce	16
2.2 MICROBIOMA INTESTINAL	21
2.2.1 Microbiota bacteriana.....	22
2.2.2 Viroma.....	25
3. OBJETIVOS	28
3.1 OBJETIVO GERAL.....	28
3.2 OBJETIVOS ESPECÍFICOS	28
4. ARTIGO CIENTÍFICO 1.....	29
5. ARTIGO CIENTÍFICO 2.....	39
6. DISCUSSÃO GERAL	70
7. CONCLUSÕES.....	72
REFERÊNCIAS	73
CURRICULUM VITAE RESUMIDO	80

LISTA DE ABREVIATURAS, SÍMBOLOS E UNIDADES

16S rRNA	Gene <i>16S Ribosomal Ribonucleic Acid</i>
ABPA	Associação Brasileira de Proteína Animal
ASAS	Do inglês <i>American Society of Animal Science</i>
DNA	Do inglês <i>Deoxyribonucleic Acid</i>
DNP	Dias Não Produtivos
DPD	Diarreia Pós-Desmame
FAO	<i>Food and Agriculture Organization</i>
HMP	Do inglês <i>Human Microbiome Project</i>
HPA	Eixo Hipotálamo-Hipófise-Adrenal
IBGE	Instituto Brasileiro de Geografia e Estatística
IDC	Intervalo Desmame-Cobertura
OECD	Do inglês <i>Organisation for Economic Co-operation and Development</i>
PIB	Produto Interno Bruto
PWD	Do inglês <i>Post-weaning diarrhea</i>
RNA	Do inglês <i>Ribonucleic Acid</i>
SCFAs	Do inglês <i>Short-Chain Fatty Acids</i>
SISCAL	Sistema Intensivo de Suínos Criados Ao Ar Livre
SISCON	Sistema Intensivo de Suínos Confinados
SNA	Sistema Nervoso Autônomo
SNC	Sistema Nervoso Central
SNE	Sistema Nervoso Entérico
USDA	Do inglês <i>United States Department of Agriculture</i>

LISTA DE FIGURAS

Figura 1. Efeitos do desmame sobre a fisiologia intestinal e a microbiota em leitões...19

Figura 2. Características e funções integradas da microbiota intestinal saudável.....21

Figura 3. Representação esquemática comparando os dois ciclos de replicação de um bacteriófago.....24

RESUMO

A fase de desmame constitui um período crítico na suinocultura tecnificada, caracterizado pela elevada vulnerabilidade dos leitões a distúrbios entéricos, como a diarreia pós-desmame (DPD), que acarreta significativas perdas econômicas. A etiologia multifatorial da DPD envolve estresse psicofisiológico, transição nutricional abrupta, imaturidade dos sistemas digestivo e imunológico e a potencial disbiose intestinal. Esta tese teve por objetivo caracterizar a microbiota bacteriana e o viroma intestinal, comparar a composição de ambos em leitões desmamados com e sem diarreia e avaliar qual a relação dessa comunidade microbiana com os distúrbios entéricos presentes nos animais estudados. Foram analisadas 30 amostras de suabes retais de leitões comerciais com 33 dias de vida, distribuídos em dois grupos: Controle (15 animais saudáveis) e Diarreia (15 animais com diarreia). A abordagem metodológica combinou sequenciamento do gene *16S rRNA* (Artigo 1) e metagenômica *shotgun* (Artigo 2) para caracterizar, respectivamente, a microbiota bacteriana e a comunidade viral intestinal. Os dados foram processados com ferramentas bioinformáticas, incluindo análises de abundância relativa, diversidade alfa e beta, abundância diferencial e, para os dados de *16S rRNA*, predição de vias metabólicas. Os resultados revelaram que leitões diarreicos apresentaram uma microbiota bacteriana enriquecida em táxons com potencial pró-inflamatórios e patogênicos, além de um enriquecimento de vias metabólicas relacionadas à degradação de compostos aromáticos (ex.: 2-aminofenol), à biossíntese de antibióticos (novobiocina) e ao metabolismo energético (ciclo do ácido tricarboxílico redutor). Paralelamente, o viroma desses animais exibiu maior abundância de vírus eucarióticos patogênicos e de bacteriófagos associados a bactérias comensais ou oportunistas. Em contraste, leitões saudáveis exibiram uma comunidade bacteriana com maior dominância de grupos com potencial probiótico e metabólico benéfico, caracterizada por um perfil enriquecido em vias de degradação de aminoácidos (ex.: L-lisina, L-treonina) e carboidratos (trealose III) e de fermentação ((S)-propano-1,2-diol), e um viroma predominado por bacteriófagos reguladores de bactérias oportunistas, além de vírus comensais. Esses achados mostram alterações significativas nas comunidades bacteriana e viral no ecossistema intestinal associado à DPD, sugerindo que o desequilíbrio nessa rede microbiana pode estar relacionado à condição clínica. Os dados podem fornecer percepções sobre os

mecanismos microbianos subjacentes à diarreia e auxiliar em estratégias profiláticas ou terapêuticas mais precisas, como o uso de moduladores microbianos, prebióticos e fagoterapia, visando à promoção da saúde intestinal dos animais.

ABSTRACT

The weaning phase is a critical period in modern pig farming, characterized by the high vulnerability of piglets to enteric disorders, such as post-weaning diarrhea (PWD), which causes significant economic losses. The multifactorial etiology of PWD involves psychophysiological stress, abrupt nutritional transition, immaturity of the digestive and immune systems, and potential intestinal dysbiosis. This thesis aimed to characterize the bacterial microbiota and intestinal virome, compare their composition in weaned piglets with and without diarrhea, and evaluate the relationship between this microbial community and the enteric disorders present in the animals studied. Thirty rectal swab samples were analyzed from commercial piglets at 33 days of age, divided into two groups: Control (15 healthy animals) and Diarrhea (15 animals with diarrhea). The methodological approach combined *16S rRNA* gene sequencing (Article 1) and shotgun metagenomics (Article 2) to characterize, respectively, the bacterial microbiota and the intestinal viral community. The data were processed with bioinformatics tools, including analyses of relative abundance, alpha and beta diversity, differential abundance, and, for *16S rRNA* data, prediction of metabolic pathways. The results revealed that piglets with diarrhea had a bacterial microbiota enriched in taxa with pro-inflammatory and pathogenic potential, as well as an enrichment of metabolic pathways related to the degradation of aromatic compounds (e.g., 2-aminophenol), antibiotic biosynthesis (novobiocin), and energy metabolism (reductive tricarboxylic acid cycle). At the same time, the virome of these animals showed a greater abundance of pathogenic eukaryotic viruses and bacteriophages associated with commensal or opportunistic bacteria. In contrast, healthy piglets exhibited a bacterial community with greater dominance of groups with beneficial probiotic and metabolic potential, characterized by a profile enriched in amino acid (e.g., L-lysine, L-threonine) and carbohydrate (trehalose III) degradation pathways, and fermentation ((S)-propane-1,2-diol), and a virome dominated by bacteriophages that regulate opportunistic bacteria, in addition to commensal viruses. These findings show significant alterations in the bacterial and viral communities in the intestinal ecosystem associated with PWD, suggesting that an imbalance in this microbial network may be related to the clinical condition. The data may provide insights into the microbial mechanisms underlying diarrhea and assist in more precise prophylactic or

therapeutic strategies, such as the use of microbial modulators, prebiotics, and phage therapy, aimed at promoting intestinal health in animals.

1. INTRODUÇÃO

A diarreia pós-desmame representa uma das fases mais críticas na suinocultura, impactando diretamente o desenvolvimento dos animais e gerando expressivas perdas econômicas (Tang et al., 2022). Essa condição apresenta etiologia multifatorial, que inclui o estresse decorrente da separação da mãe e da mistura de lotes (Rydal et al., 2023), o desmame precoce, a troca de proteína animal na dieta por proteína vegetal, e a imaturidade dos sistemas digestivo e imunológico (St-Pierre et al., 2023). Como resultado, há perda de peso, crescimento retardado e prejuízos significativos ao desempenho zootécnico e à rentabilidade do setor suinícola (Caron & Beirão, 2020).

Embora existam alguns fatores relacionados, ainda não é completamente elucidado o quadro de diarreia pós-desmame. O microbioma intestinal tem sido estudado em diferentes condições de saúde e é foco de investigações que buscam compreender sua influência sobre o desenvolvimento dos suínos durante essa fase crítica (Tancredi et al., 2025). Dessa forma, a caracterização da microbiota bacteriana e do viroma em leitões com e sem diarreia é uma abordagem para elucidar possíveis mecanismos subjacentes ao distúrbio e potencialmente subsidiar estratégias de intervenção mais eficazes. Com esse propósito, o presente estudo teve como objetivo explorar a composição da microbiota bacteriana e do viroma intestinal em suínos em fase de desmame com e sem diarreia, visando uma melhor compreensão dessas comunidades microbianas na diarreia pós-desmame.

No primeiro artigo, publicado na revista *Microbial Pathogenesis*, a microbiota bacteriana foi avaliada por meio do sequenciamento parcial do gene *16S rRNA* e análises bioinformáticas. Os resultados revelaram um perfil disbiótico nos animais diarreicos, caracterizado pelo enriquecimento de microrganismos inflamatórios e patogênicos e de vias metabólicas associadas ao metabolismo bacteriano exacerbado. Em contraste, o grupo controle apresentou uma microbiota mais equilibrada, com predomínio de grupos bacterianos de potencial probiótico e metabólico benéfico, além de bactérias comensais.

Complementando essa análise, o segundo artigo investigou o viroma dos mesmos animais, utilizando dados gerados por sequenciamento *shotgun* e ferramentas de bioinformática. Foram identificadas diferenças marcantes na comunidade viral: os

animais diarreicos apresentaram maior abundância de vírus eucarióticos patogênicos e de bacteriófagos associados a bactérias comensais ou oportunistas, enquanto o grupo saudável exibiu predominância de vírus comensais e bacteriófagos reguladores de bactérias oportunistas. Este artigo será submetido para publicação na revista *Veterinary Microbiology*.

2. REVISÃO BIBLIOGRÁFICA

2.1 SUINOCULTURA

A suinocultura constitui-se de um segmento com expressiva relevância no âmbito do mercado global de proteínas de origem animal, sendo a carne suína a segunda em termos de consumo mundial (EMBRAPA, 2024). Projeções recentes da OECD-FAO (2024) indicam uma tendência de expansão em torno de 8% no consumo global até o ano de 2033, corroborando sua significativa participação no complexo agroindustrial internacional. No contexto brasileiro, observa-se uma posição de destaque no cenário produtivo e comercial, com o país entre os maiores produtores e exportadores do mundo, segundo dados da Associação Brasileira de Proteína Animal (ABPA, 2024a).

A agroindústria brasileira demonstrou sua relevância macroeconômica em 2024 ao gerar mais de quatro milhões de empregos diretos e indiretos, contribuir com R\$ 164 bilhões para o Produto Interno Bruto (PIB) nacional e estabelecer relações comerciais com mais de 150 países (ABPA, 2024b). Dentro deste cenário, a suinocultura é um segmento estratégico para o desenvolvimento do agronegócio nacional. Em 2024, o país registrou uma produção de 5,337 milhões de toneladas de carne suína, das quais 1,308 milhões de toneladas foram destinadas para exportação, representando 24% do total produzido no país de acordo com dados da Embrapa Suínos e Aves (2024a).

O Brasil consolida sua posição no cenário global como o quarto maior produtor mundial de carne suína, respondendo por 4% da produção global, e como o terceiro maior exportador, sendo responsável por 15,7% das transações comerciais internacionais do produto (USDA, 2024; ABPA, 2025). Em relação à distribuição geográfica da produção, observa-se uma forte concentração na região Sul, com os estados do Rio Grande do Sul, Santa Catarina e Paraná respondendo por 73,13% dos abates realizados no país em 2024 (ABPA, 2025). Destacando-se o estado do Rio Grande do Sul, que ocupa a segunda posição no ranking nacional de exportações do segmento (ABPA, 2024a), reforçando sua importância na cadeia produtiva suinícola.

A suinocultura moderna passou por expressivos avanços tecnológicos nas últimas décadas, os quais vêm gerando uma reestruturação profunda dos sistemas produtivos.

Essa transformação orienta-se pela necessidade de conformidade com um paradigma tridimensional emergente, que integra critérios rigorosos de bem-estar animal, a maximização da eficiência produtiva e a adoção de práticas de sustentabilidade (Trabachini et al., 2025). A intensificação dos sistemas de produção suinícola propiciou a implementação de protocolos de monitoramento zootécnico mais precisos, acarretando expressivos ganhos de eficiência produtiva. Essa evolução resultou na minimização de perdas energéticas, na otimização da densidade animal por área produtiva, no incremento substancial dos índices de produtividade e, conseqüentemente, na significativa redução dos custos operacionais (de Castro Lippi et al., 2022).

Os sistemas de criação de suínos são classificados conforme o tipo de manejo adotado, podendo ser: sistema extensivo, sistema semiextensivo, sistema intensivo de suínos confinados (SISCON) e sistema intensivo de suínos livres (SISCAL) (Gomes et al., 2020). No sistema extensivo há nível tecnológico mais reduzido, o que diminui os custos, mas também resulta em menor produtividade. Neste modelo de criação, os animais ficam misturados e soltos em piquetes a campo, há menor controle sanitário, ambiental e alimentar, levando a crescimento lento dos animais, maior tempo de permanência nas instalações e maior risco de surgimento de doenças no plantel (Neto et al., 2023). Por essas razões, é menos adotado na suinocultura comercial, sendo mais comum em criações de pequena escala, caracterizado como suinocultura de subsistência (Faria & Almeida, 2024).

O sistema intensivo caracteriza-se pela alta densidade animal, controle ambiental rigoroso e alimentação balanceada, subdividindo-se em duas modalidades operacionais distintas. O SISCAL concentra as fases iniciais do ciclo (reprodução, gestação, maternidade e creche) em piquetes com cobertura vegetal, equipados com bebedouros, comedouros e áreas de sombreamento (Neto et al., 2023). Neste sistema a dieta é fornecida por meio de ração complementada pelo pastejo da vegetação disponível. Geralmente, ao atingir um peso vivo entre 20 e 25 kg, os leitões são comercializados para outras unidades produtivas, onde as fases subseqüentes de crescimento e terminação são realizadas em sistemas de confinamento, que permitem um manejo sanitário e nutricional mais rigoroso (Faria & Almeida, 2024).

O SISCON é conduzido integralmente em instalações fechadas e climatizadas, onde os animais permanecem alojados em baias especializadas por categoria (Fortes, 2022). Todo o ciclo produtivo ocorre em ambiente controlado, com alimentação específica para cada fase e o emprego de genética de alto potencial para conversão alimentar eficiente (Echeverria, 2022). Seus benefícios incluem alta produtividade, melhoramento genético sistemático e eficiência de área, mas o sistema enfrenta desafios como custos elevados, impacto ambiental pela concentração de dejetos e questões relacionadas ao bem-estar animal, demandando manejo especializado para mitigação (Faria & Almeida, 2024). Em suma, cada fase da produção apresenta particularidades relacionadas à nutrição, manejo, instalações e cuidados específicos, cabendo ao suinocultor a escolha do sistema com base em seu nível de tecnificação, experiência e objetivos produtivos.

Conforme estabelecido pela Portaria SEEC N° 146 (Portaria SEEC, 2025), as unidades produtivas voltadas à suinocultura são classificadas de acordo com suas especificidades operacionais e sanitárias. Granjas de Reprodutores Suínos Certificadas (GRSC) correspondem àquelas que obedecem rigorosamente a todos os critérios exigidos para a certificação em biossegurança avançada, sendo reconhecida como área livre das enfermidades. Já a Unidade Produtora de Leitões (UPL) caracteriza-se por ser especializada na obtenção de leitões, os quais, logo após o desmame, são destinados à engorda em outro local, abrangendo as fases de cobertura, gestação, maternidade, creche e, eventualmente, uma central de inseminação de uso restrito. O crechário ou creche (CR) consiste no local que acolhe os leitões recém-desmamados oriundos da UPL, mantendo-os até o final da fase de creche ou até que estejam aptos ao abate. A Terminação, por outro lado, é a instalação responsável pela engorda dos suínos, preparando-os para o encaminhamento ao frigorífico, e há o sistema de Ciclo Completo (CC), no qual todas as fases de produção são realizadas em um mesmo estabelecimento em regime contínuo (Portaria SEEC, 2025).

2.1.1 Desmame precoce

A suinocultura atual adota o desmame precoce entre duas e quatro semanas de idade, significativamente antes do período natural que ocorreria entre 12 e 18 semanas na

natureza (St-Pierre et al., 2023). Esta prática, embora economicamente vantajosa, apresenta desafios consideráveis devido à imaturidade fisiológica dos leitões (Qin et al., 2024). Os sistemas digestivo e imunológico ainda em desenvolvimento, combinados com a interrupção abrupta do fornecimento de imunoglobulinas e outros componentes bioativos do leite materno, resultam em maior suscetibilidade a patógenos e distúrbios intestinais (St-Pierre et al., 2023).

Um estudo de 2024 analisou os avanços recentes na pesquisa em suinocultura com base nos trabalhos apresentados em duas conferências internacionais: a reunião anual da *American Society of Animal Science* (ASAS), realizada em Albuquerque (NM, EUA) em julho de 2023, e a conferência da seção Centro-Oeste, em Madison (WI, EUA) em março do mesmo ano. Dos 295 trabalhos apresentados nessas conferências, 47% concentraram-se em leitões de creche, demonstrando um direcionamento significativo das pesquisas em nutrição e produção para esta fase crítica do desenvolvimento (Kim et al., 2024). Esse foco prioritário reflete as complexidades fisiológicas inerentes ao período pós-desmame, uma vez que os leitões são desmamados precocemente, com o trato gastrointestinal ainda em desenvolvimento e com capacidade limitada para processar os alimentos convencionais fornecidos nessa fase (Deng et al., 2023).

Na indústria suinícola, o desmame precoce é uma prática zootécnica consolidada para aumentar a eficiência reprodutiva do rebanho. Sua eficácia é demonstrada pela redução significativa do intervalo entre partos, diminuindo diretamente os dias não produtivos (DNP) da matriz, permitindo um maior número de ciclos por ano (Qiao et al., 2023). Embora o retorno ao estro ocorra comumente em torno de 7 dias, evidências indicam que linhagens específicas, como fêmeas mestiças de Landrace e Large White, podem manifestar o estro precocemente, em aproximadamente 5 dias após o desmame, otimizando ainda mais o intervalo entre partos e a produtividade do plantel (Gianluppi et al., 2020). Estudos anteriores demonstram que o intervalo desmame-cobertura (IDC) ideal situa-se entre 4 e 6 dias, período no qual se observa maior taxa de parto e maior número de leitões nascidos vivos na ninhada subsequente, em comparação com IDC mais longos (7 a 10 dias), que resultam em índices reprodutivos significativamente inferiores (Nam & Sukon, 2025). Pesquisas recentes mencionam que a implementação do desmame precoce pode ser uma estratégia para reduzir significativamente a transmissão vertical de

patógenos excretados pela matriz durante a lactação, limitando a exposição dos leitões a esses agentes infecciosos (Metallo et al., 2025).

De acordo com a Diretiva 2008/120/EC, países da União Europeia como Dinamarca, Holanda e Alemanha permitem o desmame de leitões aos 28 dias, ou aos 21 dias em sistemas de produção de partos em lote. A Suécia, no entanto, se destaca por proibir a prática em leitões com menos de quatro semanas de vida (de Castro Lippi et al., 2022). Este período, independentemente da genética do plantel, determina de forma mais crucial os índices de conversão alimentar e o desenvolvimento dos leitões, superando em influência as condições de aleitamento (Mahmud et al., 2023).

O desmame constitui uma transição crítica na criação de suínos, caracterizada por mudanças abruptas e multifatoriais que geram elevado estresse aos leitões. Nesta fase, os animais são submetidos à separação materna, à mistura de lotes com estabelecimento de nova hierarquia social e à transferência para um ambiente desconhecido (Metallo et al., 2025). Simultaneamente, ocorre uma drástica transição nutricional, na qual a dieta líquida e imunologicamente rica em nutrientes de origem animal (leite), é substituída por uma ração sólida de origem vegetal, desprovida de tais componentes bioativos (Rivera et al., 2025). Nessa fase, os leitões possuem imaturidade fisiológica, na qual os sistemas digestivo e imunológico ainda não estão completamente desenvolvidos. A abrupta privação do leite materno, fonte essencial de imunoglobulina A e fatores de proteção, elimina a imunidade passiva, o que aumenta a suscetibilidade a patógenos e precipita a ocorrência de distúrbios e doenças gastrointestinais no período pós-desmame (Figura 1) (St-Pierre et al., 2023).

O desmame precoce exacerba o estresse nos leitões, podendo comprometer a função da barreira epitelial intestinal, o que aumenta significativamente o risco de disfunções entéricas (Figura 1) (Tang et al., 2022). Estas alterações levam ao acúmulo de líquido no lúmen intestinal, resultando frequentemente em quadros de diarreia que podem evoluir para casos graves com desidratação e mortalidade (Tao et al., 2022). Essa sucessão de desafios induz a ativação do eixo hipotálamo-hipófise-adrenal (HPA), culminando na elevação aguda das concentrações de cortisol e causando imunossupressão. Esta resposta, observada independentemente da idade de desmame e

amplamente documentada na literatura, serve como um marcador biológico consagrado para o estado de estresse fisiológico e psicológico característico da fase pós-desmame (Van Kerschaver et al., 2023).

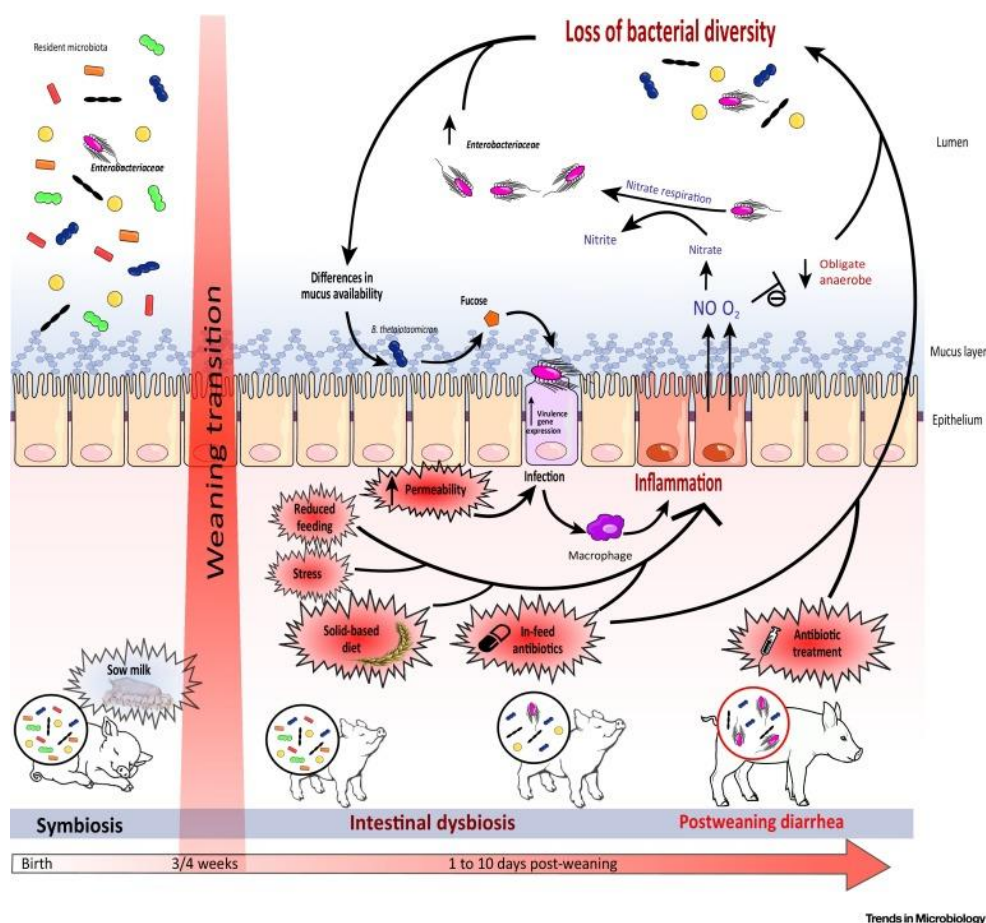


Figura 1. Efeitos do desmame sobre a fisiologia intestinal e a microbiota bacteriana intestinal em leitões. A figura ilustra os processos que levam à perda de diversidade bacteriana intestinal causando a diarreia pós-desmame. Os principais fatores que causam perturbação mostrados são: introdução de dieta sólida e tratamento com antimicrobianos, que levam a redução de bactérias anaeróbias benéficas e a alterações na disponibilidade de substratos como o nitrato e a fucose. O comprometimento da barreira intestinal e a ativação da resposta imune inflamatória completam um ciclo que desencadeia o quadro clínico de diarreia. O modelo integra assim componentes microbianos, dietéticos e do hospedeiro para explicar a patogênese da doença (Reprodução autorizada, Gresse et al., 2017).

2.1.2 Doenças entéricas em suínos

As doenças entéricas representam um dos maiores desafios sanitários e econômicos enfrentados pela suinocultura em escala global, ocasionando desde prejuízos no desempenho zootécnico dos animais até a ocorrência de óbitos (Ucero-Carretón et al., 2026). A colibacilose, causada por cepas patogênicas de *Escherichia coli*, sendo o patotipo mais relevante no suíno a *E. coli enterotoxigênica (ETEC)*, incide predominantemente no período neonatal e após o desmame (Pokharel et al., 2023). A diarreia neonatal ocorre de 1 a 4 dias após o parto, já a diarreia pós-desmame geralmente aparece de 2 a 3 semanas após o desmame e é caracterizada pela aparência de diarreia aquosa, que pode variar de amarelo-acinzentado ou levemente rosada, durando até uma semana (Barros et al., 2023; Sukalić et al., 2026). Em leitões neonatos, infecções por *Clostridium perfringens* tipo C estão entre as principais causas de enterite grave, caracterizada por enterite necrosante e alta taxa de mortalidade (Uzal et al., 2023). A doença apresenta ampla distribuição nos sistemas de produção de suínos e acarreta impactos econômicos significativos, sobretudo devido às altas taxas de mortalidade que podem ocorrer nos animais infectados (Posthaus et al., 2020). Já *Clostridium perfringens* tipo A tem sido associado à diarreia em leitões na primeira semana de vida, embora sua ocorrência já tenha sido descrita também em leitões saudáveis (Ambrós et al., 2025).

Os vírus da diarreia epidêmica suína (PEDV) e da gastroenterite transmissível (TGEV) são coronavírus entéricos pertencentes ao gênero *Alphacoronavirus*, responsáveis por causar diarreia aguda e grave em suínos de todas as idades, com impacto maior em leitões neonatos (Yan et al., 2026). Em leitões, ambos os agentes levam a altas taxas de morbidade e mortalidade, devido à desidratação grave, atrofia das vilosidades intestinais e má absorção. A transmissão ocorre pela via fecal-oral, através de fômites e contato direto (Weng et al., 2026).

Outra virose importante, a rotavirose, causada por rotavírus dos grupos A, B e C, é uma enfermidade viral que acomete leitões, caracterizada por diarreia aquosa, desidratação e perda de peso, podendo evoluir para óbito em casos mais graves (Jacobson, 2022). O rotavírus suínos A, mais associado ao período pós-desmame, é responsável por

infecções entéricas em leitões entre 1 e 8 semanas de idade (Jampanil et al., 2025). A doença apresenta alta morbidade e impacto econômico na suinocultura, sendo de difícil controle devido à elevada resistência do vírus no ambiente, o que favorece sua persistência e disseminação (Rybkowska et al., 2025).

Outra causa de diarreia em leitões são infecções por *Cystoisospora suis*, patógeno responsável por causar diarreia e desidratação em animais de 2 a 4 semanas de idade. A infecção leva à diminuição das vilosidades intestinais e redução do ganho de peso corporal, efeitos que persistem desde a fase clínica até o pós-desmame (6–7 semanas de idade) (Han et al., 2025). A diversidade de agentes envolvidos nas doenças entéricas de leitões mostra que os quadros de diarreia raramente têm uma única causa. Estudar de forma integrada essas múltiplas etiologias é essencial para o diagnóstico preciso, a prevenção eficaz e a redução das perdas produtivas, garantindo assim a sustentabilidade da suinocultura.

2.2 MICROBIOMA INTESTINAL

O trato gastrointestinal abriga o microbioma mais complexo do organismo, um ecossistema formado pela comunidade de microrganismos, seu material genético, seus metabólitos e o ambiente intestinal. Esta interação resulta na maior diversidade de espécies e densidade celular do corpo (St-Pierre et al., 2023). Localizado na interface da mucosa intestinal, este microbioma desempenha funções fundamentais para a fisiologia do hospedeiro, incluindo a manutenção da homeostase do epitélio intestinal, o auxílio na extração e utilização de energia dos nutrientes, a modulação e maturação do sistema imunológico (Li et al., 2024).

Nas últimas décadas, o interesse científico pelo microbioma expandiu-se de forma notável, culminando na criação de iniciativas de grande escala. O marco mais proeminente foi o Projeto do Microbioma Humano (*Human Microbiome Project* - HMP), lançado em 2007, que realizou o primeiro mapeamento e caracterização sistemática da vasta comunidade microbiana intestinal humana (Turnbaugh et al., 2007). A contribuição do HMP para a base de conhecimento da área foi substancial, com seus pesquisadores publicando mais de 650 artigos científicos até o final de 2017 (Human Microbiome

Project (HMP) (NIH Common Fund, 2025). Essa mobilização de esforços consolidou o microbioma como um campo prioritário de pesquisa, conferindo-lhe atenção crescente e acelerando exponencialmente a produção de conhecimento sobre sua complexa estrutura e multifuncionalidade (Shah et al., 2025). Como resultado, o estudo do microbioma transformou-se em uma área de investigação fundamental e transversal nas ciências biológicas e da saúde.

O microbioma intestinal passa por diversas alterações desde o nascimento do indivíduo. Essas alterações são influenciadas por diversos fatores endógenos e exógenos, como idade, genética, dieta, manejo, estresse, antibioticoterapia (Passos, 2024). Entre as formas de analisar as alterações no microbioma intestinal está a razão *Bacillota/Bacteroidota*, já que alguns estudos apontam essa relação como potencial biomarcador para a saúde intestinal em animais e humanos (Karačić et al., 2024). Esses filos juntamente com *Pseudomonadota* estão entre os principais constituintes do intestino suíno independentemente da idade (Wedemeyer et al., 2025).

2.2.1 Microbiota bacteriana

A microbiota intestinal constitui o componente vivo e metabolicamente ativo do microbioma, sendo composta por grupos taxonômicos fundamentais – bactérias, arqueias, fungos e vírus – que estão presentes ao longo de toda a vida do hospedeiro, variando as proporções relativas destes grupos e a diversidade de espécies ao longo do tempo (Van Hul et al., 2024). É por meio da atividade integrada desta comunidade e pela produção de metabólitos que uma série de funções fisiológicas essenciais do hospedeiro são reguladas, incluindo o metabolismo de nutrientes, o desenvolvimento e homeostase dos tecidos, a modulação do sistema imunológico e a proteção contra a colonização por patógenos (Leonard & Toro, 2023)

A produção de compostos bioativos pela microbiota intestinal, particularmente os ácidos graxos de cadeia curta (SCFAs - *short chain fatty acids*), regula importantes funções homeostáticas, promovendo a integridade da barreira epitelial e exercendo atividade anti-inflamatória (Kim & Mills, 2024). Fatores como dieta, uso de antibióticos, níveis de estresse e exposição a patógenos modulam decisivamente a composição desta

comunidade microbiana, posicionando-a como um regulador central da imunidade do hospedeiro (Figura 2) (Kim & Mills, 2024; Van Hul et al., 2024)

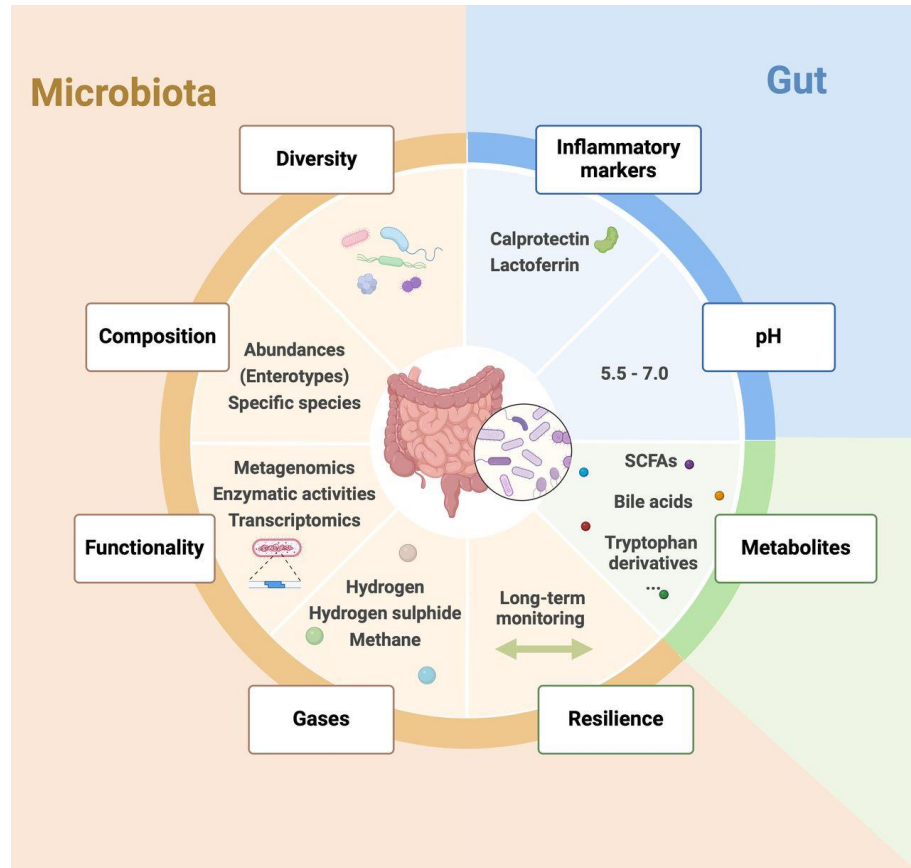


Figura 2. Características e funções integradas da microbiota intestinal saudável. O modelo mostra alguns atributos inter-relacionados: diversidade microbiana (riqueza e equilíbrio da comunidade); abundância de espécies específicas (perfil taxonômico); espectro de funções metabólicas; produção de gases (ex.: hidrogênio e metano); resiliência (capacidade de retorno à homeostase após perturbações); produção de metabólitos bioativos (ex.: SCFAs e ácidos biliares); modulação do pH intestinal; e influência sobre marcadores inflamatórios locais e sistêmicos. O diagrama evidencia a natureza multifatorial e interconectada dessas características na definição da estrutura, função e impacto da microbiota na saúde intestinal. (Reprodução autorizada, Van Hul et al., 2024).

Para manter a homeostase intestinal e prevenir patologias, a microbiota bacteriana utiliza mecanismos como competição ecológica por nutrientes e sítios de adesão, produção de metabólitos antimicrobianos e modulação imunológica (Figura 2), conferindo proteção contra enteropatias. Cepas de bactérias com propriedades probióticas, como *Lactobacillus* e *Bifidobacterium*, exploram esses mecanismos e demonstram eficácia profilática contra diarreias em suínos (Lv et al., 2025).

O eixo intestino-cérebro é um sistema de comunicação bidirecional no qual disfunções gastrointestinais e neurológicas se influenciam mutuamente por meio de mecanismos neuroimunes e neuroendócrinos. No contexto do desmame suíno, esse fenômeno manifesta-se claramente quando o estresse social decorrente da mistura de lotes desencadeia alterações neurogênicas que comprometem a homeostase intestinal, resultando em quadros diarreicos (Savulescu-Fiedler et al., 2025).

Esse eixo constitui um sistema neuro-humoral complexo, fundamental para a manutenção da homeostase metabólica do organismo e integra múltiplos componentes: os sistemas nervosos central (SNC) e entérico (SNE), o sistema nervoso autônomo (SNA), com seus ramos simpático e parassimpático, e sistemas neuroendócrinos e imunológicos funcionalmente interligados (Hemmings, 2022). A comunicação ocorre por três vias principais: endócrina, mediada por hormônios; nervosa, transmitida sobretudo pelos ramos do nervo vago; e imunológica, efetuada por moléculas sinalizadoras como as citocinas. Nesta última, destacam-se as citocinas pró-inflamatórias, capazes de ativar o eixo hipotálamo-hipófise-suprarrenal (HPA) e induzir a secreção de cortisol, o que pode exacerbar a permeabilidade intestinal e potencializar um ciclo de desregulação e inflamação sistêmica (Passos, 2025).

O estresse agudo atua nessa rede multifacetada, alterando a microbiota intestinal por meio da ativação do SNA e do eixo HPA, além de influenciar diretamente o sistema nervoso entérico, as células enteroendócrinas e as células do sistema imune. Bactérias intestinais possuem receptores para neurotransmissores que, quando ativados, modificam seu comportamento funcional (Leigh et al., 2023). Paralelamente, o estresse aumenta a permeabilidade intestinal, altera a composição do muco, induz degranulação mastocitária – com consequente disfunção muscular intestinal – e estimula a expressão de fatores de

virulência em patógenos como *Pseudomonas aeruginosa* e *Campylobacter jejuni* (Savulescu-Fiedler et al., 2025).

2.2.2 Viroma

O viroma intestinal constitui uma das comunidades virais mais complexas documentadas na natureza, sendo predominantemente composto por bacteriófagos (nas formas livre e integrada), vírus eucarióticos e viroides (Chica Cardenas et al., 2025). Esta comunidade exibe notável diversidade genômica, abrangendo vírus de DNA e RNA de fita simples e dupla, organizados em um espectro taxonômico que inclui desde partículas que infectam células do hospedeiro até bacteriófagos especializados em parasitar comunidades bacterianas, além de vírus de origem ambiental incorporados por meio da dieta (Tun et al., 2024).

A caracterização deste ecossistema enfrenta obstáculos metodológicos significativos, principalmente por não haver marcadores virais específicos e por haver grande diversidade genética viral. Não obstante essas limitações, pesquisas recentes demonstram que o viroma intestinal exerce funções regulatórias cruciais na homeostase microbiana, modulação imunológica e processos metabólicos, impactando diretamente a fisiologia do hospedeiro (Tun et al., 2024; Zeng et al., 2024). A composição do viroma intestinal, assim como a composição da microbiota bacteriana intestinal, muda de acordo com o ambiente e estado fisiológico do hospedeiro (Ritz et al., 2024).

Os bacteriófagos são as entidades biológicas mais abundantes e diversificadas nos ecossistemas microbianos. Estimativas apontam a existência de aproximadamente 10^{31} partículas bacteriofágicas na biosfera, um número que supera o total de células procarióticas (Loponte et al., 2021). Essa dominância quantitativa está intrinsecamente ligada a um papel ecológico fundamental: os bacteriófagos são reguladores cruciais das populações bacterianas, atuando como um dos principais mecanismos na manutenção do equilíbrio e na dinâmica das comunidades microbianas (Y. Ji et al., 2024).

A atividade biológica dos bacteriófagos é regida por dois principais ciclos de replicação: o ciclo lítico, no qual bacteriófagos virulentos replicam-se ativamente e lisam a célula hospedeira para liberar novos vírions (Figura 3A); e o ciclo lisogênico, onde

bacteriófagos temperados integram seu genoma no cromossomo bacteriano, permanecendo como profagos (temperados) que se replicam passivamente durante a divisão celular da bactéria (Figura 3B) (Chen et al., 2025). Após a infecção, bacteriófagos temperados podem seguir o ciclo lítico ou lisogênico, a definição entre esses destinos é determinada por condições ambientais, como a densidade da população bacteriana e a disponibilidade de nutrientes (Figura 3B) (Gummalla et al., 2023).

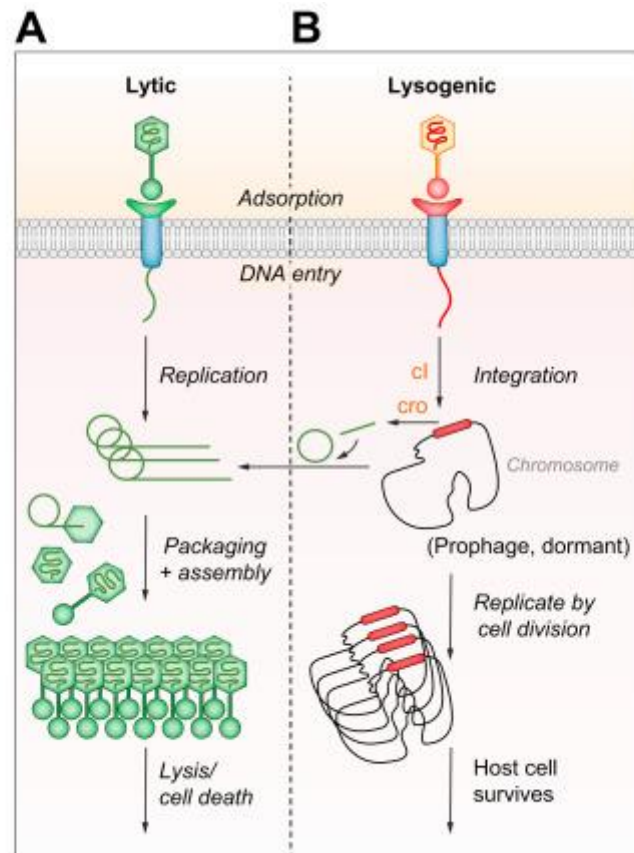


Figura 3. Representação esquemática comparando os dois ciclos de vida de um bacteriófago.

No ciclo lítico (A), o vírus invade a célula, replica seu material genético, produz novas partículas virais e causa a lise (morte) da célula hospedeira. No ciclo lisogênico (B), o material genético viral é integrado ao cromossomo bacteriano (formando um profago), se replica passivamente junto com a célula hospedeira e pode permanecer latente por várias gerações, sem causar a morte imediata da célula. Sob certas condições, o profago pode ativar-se e entrar no ciclo lítico (Reprodução autorizada, Hsu et al., 2021).

Um estudo com camundongos observou que o aumento de bacteriófagos da classe *Caudoviricetes* (anteriormente *Caudovirales*) melhorou as funções de

reconhecimento e memória. Esse resultado apoia a hipótese de que o viroma intestinal pode modular o comportamento e a cognição do hospedeiro por meio de interações com a microbiota intestinal e do eixo intestino-cérebro (Mayneris-Perxachs et al., 2022).

Apesar do conhecimento crescente sobre o impacto do microbioma intestinal em funções fisiológicas e patológicas, o conhecimento sobre o viroma intestinal e, sobretudo, sobre a interação entre bactérias e vírus no contexto da DPD ainda é limitado. A compreensão integrada dessas comunidades microbianas é fundamental para elucidar os mecanismos subjacentes à disbiose intestinal e à manifestação clínica da diarreia no período pós-desmame. Nesse contexto, a caracterização da microbiota bacteriana e do viroma intestinal de leitões com e sem diarreia, por meio de abordagens de sequenciamento de alto desempenho, tem potencial em identificar assinaturas microbianas associadas à saúde ou à doença, contribuindo para o desenvolvimento de estratégias profiláticas e terapêuticas mais direcionadas, sustentáveis e alinhadas às demandas atuais da produção animal.

3. OBJETIVOS

3.1 OBJETIVO GERAL

Investigar o papel da microbiota bacteriana e viral intestinal na ocorrência de diarreia em leitões durante a fase de desmame, contribuindo para o entendimento dos fatores microbianos associados à saúde intestinal.

3.2 OBJETIVOS ESPECÍFICOS

Caracterizar a composição da microbiota bacteriana e da comunidade viral intestinal de leitões com e sem diarreia;

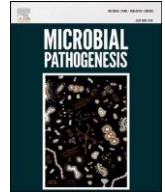
Descrever e comparar a diversidade microbiana intestinal entre leitões sadios e com diarreia na fase de desmame;

Avaliar as vias metabólicas bacterianas intestinais e sua possível associação com a ocorrência de diarreia;

Investigar a relação entre o perfil microbiano (bacteriano e viral) e distúrbios entéricos em leitões desmamados.

4. ARTIGO CIENTÍFICO 1

O primeiro artigo desta tese teve por objetivo caracterizar e comparar a microbiota bacteriana entre leitões desmamados com e sem diarreia pós-desmame. O artigo está publicado na revista científica *Microbial Pathogenesis*.



Impact of weaning on piglet microbiota: differences in bacterial taxa and metabolic pathways associated with diarrhea

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ARTICLE INFO

Keywords:

Metataxonomics
16S rRNA
Gastrointestinal microbiome
Diarrhea
Stress weaning

ABSTRACT

In modern pig farming, early weaning increases pigs' vulnerability to intestinal diseases due to the immaturity of their digestive and immune systems. Post-weaning diarrhea is a common issue in Brazil, leading to weight loss and slow growth, which significantly impacts the swine industry's performance. The intestinal microbiota plays a crucial role during this phase, as it is determinant for the animals' health and productivity. This study aimed to compare the microbiota of post-weaning piglets with and without diarrhea, identifying differences in the bacterial composition and microbial metabolic pathways. The most frequent phyla in both groups were *Bacillota* followed by *Bacterioidota* and *Pseudomonadota*, which together accounted for nearly 90% of microbiota. However, one phylum and eight bacterial families were enriched in the control group (minimum fold change of 4.6), whereas only the *Fusobacteriaceae* family was enriched in diarrhea group (fold change of 6.5). Moreover, 16 genera were enriched in the control group, while 12 were enriched in the diarrhea group. These findings indicate distinct microbiota compositions between the groups, suggesting an association with dysbiosis. Variations in metabolic pathways were also observed. In the diarrhea group, six pathways were enriched (fold changes of 8.12–263.23), most of which were associated with pathogenic and inflammatory processes. In addition, an increase in pathways linked to bacterial metabolism suggested heightened microbial activity, potentially exacerbating diarrhea. In contrast, seven metabolic pathways were enriched in the control group, many of which are essential for immune system development and strengthening, supporting healthy growth. These results highlight key differences in the microbiota and metabolic activity of piglets with and without diarrhea, providing insights that may aid in developing strategies to promote intestinal health and improve post-weaning performance.

1. Introduction

In natural conditions, pigs are weaned between 12 and 18 weeks of age. However, in modern swine farming, weaning occurs much earlier, typically between two and four weeks of age, increasing the challenges associated with this transition. At this stage, piglets have immature digestive and immune systems. The abrupt loss of immunoglobulins and other components provided by milk further increases their susceptibility

to pathogens, making them more prone to gastrointestinal disorders and diseases [1].

Weaning is recognized globally as one of the most critical and stressful periods in pig farming, with significant impacts on the animals' development and health. A Chinese study of 120 pigs found that weaning stress impacted all individuals, regardless of breed or gender [2]. A Finnish study of 170 female pigs reported that post-weaning microbiota had a greater influence on growth and weight gain than

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<https://doi.org/10.1016/j.micpath.2025.107878>

Received 1 November 2024; Received in revised form 11 March 2025; Accepted 3 July 2025

Available online 4 July 2025

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the breastfeeding period, emphasizing the importance of this phase [3]. Within the first two weeks after weaning, piglets may develop severe diarrhea, leading to dehydration and, in some cases, death. This is a problem affecting swine industry worldwide, including Brazil, where post-weaning diarrhea is prevalent in most farms during this critical phase [4].

Post-weaning diarrhea can lead to intestinal dysbiosis, driven by dietary, social, and environmental changes that disrupt microbiota balance and weaken the piglets' immune system [5]. During this phase, piglets exhibit several physiological limitations, including insufficient production of digestive enzymes, reduced nutrient absorption due to villous underdevelopment, immature gastric secretion, and low feed intake. These factors contribute to the growth limitations commonly observed during this period [6].

In addition, piglets become more susceptible to infections, promoting the proliferation of pathogenic bacteria such as *Escherichia coli*, *Clostridium perfringens*, *Salmonella choleraesuis*, and *Salmonella enterica* subsp. *enterica* serovar Typhimurium. At the same time, there is a reduction in the populations of beneficial bacteria, such as *Lactobacillus sobrius*, *L. acidophilus*, and *L. reuteri* (Silva et al., 2018). These alterations can have significant consequences, including weight loss and impaired growth, ultimately affecting swine production efficiency [7].

The intestinal microbiota plays an essential role in food degradation and fermentation, as well as in pathogen defense through competition for nutrients and adhesion sites and the secretion of antimicrobial peptides [8]. This protective function, known as colonization resistance, helps shield the host from infections [9]. The microbiota composition is influenced by various factors and differs under different management conditions. Understanding the pig microbiota and its diversity not only may enhance production quality but also provide insight into the determinants of health and disease in pigs and other species they interact with, including humans [10]. Studies on post-weaning diarrhea and its association with specific microorganisms, such as *Escherichia coli* and *Salmonella* Typhimurium, have been conducted in multiple countries [11,12]. In Brazil, research on weaned pigs has primarily focused on animal performance and stress responses, particularly in states like Santa Catarina and São Paulo [13,14].

Investigating animal microbiota across different production systems and regions is essential for understanding its role in post-weaning diarrhea. In Brazil, few studies have examined this condition in field-raised pigs, and none have focused on animals from Rio Grande do Sul. This study aimed to identify microbial taxa associated with intestinal dysbiosis in weaned piglets on a farm in this state. By analyzing fecal samples, we compared the microbial structure of piglets with diarrhea to that of healthy counterparts, providing a detailed insight into fecal microbiota composition during this critical phase.

2. Materials and methods

2.1. Ethics Committee and approval

This study was approved by the Ethics Committee for the Use of Animals of the Federal University of Rio Grande do Sul (CEUA-UFRGS), under approval number 33558/2018. All procedures involving the animals were carried out in accordance with animal welfare standards, aiming to minimize any discomfort and suffering during the experiment.

2.2. Experimental design and sample collection

Samples from post-weaning pigs with diarrhea ($n = 15$, diarrhea group) and pigs without diarrhea ($n = 15$, control group) were obtained by convenience. Diarrhea was determined by the presence of liquid stools and the number of samples was defined based on the number of piglets with diarrhea at the time of collection ($n = 15$). To ensure equivalence between the groups, the same number of samples was randomly collected from healthy piglets. The diarrhea piglets were all

housed in the same pen, while the healthy piglets were housed in 10 animals/pen. The animals were from Duroc breed and in both groups, were 33 days old and in the nursery phase. The animals were sourced from a Certified Swine Breeding Farms (CSBF) in Rio Grande do Sul, south Brazil.

In the transition phase from milk to solid food (around 15 days of age), the animals received a pre-starter feed. Although we did not had access to exact feed formula, this kind of feed in Brazil usually contains highly digestible ingredients like fish meal, powdered milk, soy, and corn, along with vitamins, minerals, and additives for intestinal health and weight gain. Probiotics, prebiotics, and antioxidants are also included to support the microbiota and immune system.

At weaning, 28-day-old piglets were subjected to the following immunization plan: autogenous vaccine against *Actinobacillus pleuropneumoniae* (Microvet, Brazil), M + PAC against *Mycoplasma hyopneumoniae* (MSD Animal Health, USA), and Circumvent PCV against porcine circovirus type 2 (PCV2; MSD Animal Health, USA). In addition, as part of farm management, all animals received antibiotic treatment with Norfloxacin 14 mg/kg (Farmaflox®, Farmabase™, Brazil) added to the drinking water at 21 days for seven consecutive days and Tula-thromycin a single dose of 2.5 mg/kg of body weight administered intramuscularly (Draxin®, Zoetis™, USA).

Piglets that exhibited diarrhea symptoms were housed in a separate pen from the other animals. To avoid environmental contamination and for the sake of animal welfare, rectal swabs were collected from the pigs with clinical signs of diarrhea ($n = 15$) and from animals that did not have diarrhea ($n = 15$). The swabs were placed in sterile tubes containing 400 µl of RNALater® (ThermoFisher Scientific, USA). The swabs were transported from the farm to the laboratory on ice on the same day of collection and subsequently stored at -70°C until sample processing.

2.3. DNA extraction and 16S rRNA partial gene amplification

Before DNA extraction, the samples were homogenized using a Precellys®24 (Bertin Technologies S.A.S, France). Microbial DNA was extracted using the MagMax Pathogen DNA/RNA kit (Thermo Fisher Scientific, USA) following the manufacturer's instructions. The V4 region of the bacterial 16S rRNA gene was amplified by PCR (94°C for 2 min, followed by 35 cycles of 94°C for 30 s, 55°C for 30 s, and 68°C for 45 s, with a final extension at 68°C for 5 min) using 16S-forward ($5'$ TCG TCG GCA GCG TCA GAT GTG TAT AAG AGA CAG GTG CCA GCM GCC GCC GCG GTA $3'$) and reverse ($5'$ GTC TCG TGG GCT CGG AGT TGT GTA TAA GAG ACA GGG ACT ACH VGG GTW TCT AAT $3'$) primers, which contain the Illumina adapter sequence (Kozich et al., 2013). The reactions were performed using Phusion polymerase and 12.5 ng of DNA, following the standard protocol for this enzyme. The amplicons were purified using Agencourt AMPure XP magnetic beads (Beckman Coulter, USA) and indexed according to the Illumina protocol. The libraries were quantified using a Qubit 2.0 fluorometer (Invitrogen, USA), and the concentrations were normalized for sequencing, which was performed using the MiSeq Reagent Kit v2 500 cycles (2×250 paired-end) on the MiSeq sequencer (Illumina, USA).

2.4. Bioinformatics analysis

The sequence data was processed using a pipeline from CLC Genomics Workbench software version 23.0.4 (QIAGEN, DEU), following manufacturer instructions for microbiome analysis. Initially, the sequences underwent quality analysis and were trimmed to remove indexes and adapters. A quality filter with a Phred score lower than 20 was used to eliminate low-quality reads, and potentially chimeric sequences were identified and removed.

After the initial filtering and trimming steps, the remaining sequences were clustered into Operational Taxonomic Units (OTUs) based on a 99 % identity level and classified against the SILVA v138 reference database [15]. Microbial communities were analyzed, in which alpha

diversity, beta diversity, taxonomic distribution, and differential abundance analyses were conducted. Functional profiling was analyzed using PICRUST2 software [16] with a MetaCyc database [17] which has a broader and more complete set of metabolic pathway data [18].

2.5. Statistical analyses

The microbial alpha diversity of the samples was determined using abundance-based coverage estimation (ACE), Chao1 diversity index, Faith's phylogenetic diversity (PD), Simpson index, and Shannon index. For statistical comparisons of these indexes, a Mann Whitney test was applied. Beta diversity indices of the microbial samples were determined using principal coordinate analysis (PCoA) based on Bray-Curtis dissimilarity, UniFrac phylogenetic diversity, and Jaccard similarity index. To obtain statistical confidence in the observed sample clustering by PCoA, permutation multivariate analysis of variance (PERMANOVA) was performed on the distance matrix. For differential abundance, a generalized linear model (GLM) was used to analyze differences in feature abundance (taxa or metabolic pathways) across sample groups. It assumes Negative Binomial distribution for abundances and applied the Likelihood Ratio test for group-wide comparisons. Features with $P < 0.05$ were considered enriched.

3. Results

3.1. Quality analysis and trimming

The sequences were imported into the CLC platform, where the quality analysis of the 30 samples was performed, followed by trimming. Two sequences from the diarrhea group were removed due to a small number of reads after trimming (5D: 48,128 and 11D: 800). The remaining sequences were then composed of 15 from the control group and 13 from the diarrhea group, totaling 28 sequences (Supplementary Table 1).

3.2. Taxonomic analysis

A total of 40,335 bacterial taxa (OTUs) belonging to 27 phyla, 157 families, 397 genera and 634 species were identified (Supplementary Table 2). In addition, 142 genera and 948 species were identified as nonclassified taxa. The most frequent phyla were *Bacillota* (57 % and 65 %), *Bacteroidota* (17 % and 16 %), *Pseudomonadota* (14 % and 7 %) in the control and diarrhea groups, respectively, and *Campylobacteriota* (6 %) in both groups. The most abundant families were *Lactobacillaceae*, *Lachnospiraceae*, *Prevotellaceae*, *Veillonellaceae*, *Ruminococcaceae*, and *Oscillospiraceae* in both groups (Fig. 1). The ratio between *Bacillota*/*Bacteroidota* was 3.35 in the control group and 4.06 in the diarrhea group, which may indicate dysbiosis.

3.3. Microbial diversity

There were no statistically significant differences between the control and diarrhea groups in any of the evaluated indices: Chao1 ($p = 0.525$), Simpson ($p = 0.717$), and Shannon ($p = 0.683$) (Fig. 2A). Principal coordinates analysis (PCoA) based on Bray-Curtis dissimilarity ($p = 0.353$; Fig. 2B), Jaccard ($p = 0.414$), weighted UniFrac ($p = 0.510$), and unweighted UniFrac ($p = 0.390$) also indicated no statistically significant differences between the groups.

3.4. Differential abundance analysis

Only one enriched phylum was found, *Patescibacteria*, in the control group. The families enriched in the control group included *Acetobacteraceae*, *Pseudomonadaceae*, *Saccharimonadaceae*, *Erwiniaceae*, *Alteromonadaceae*, *Mycobacteriaceae*, *Carnobacteriaceae*, *Enterobacteriaceae*, and only the family *Fusobacteriaceae* in the diarrhea group (Fig. 3A). Regarding the genera, in the diarrhea group, the differentially abundant genera were *Rheinheimera*, *Candidatus*, *Saccharimonas*, *Pantoea*, *Massilia*, *Pseudomonas*, *Gluconacetobacter*, *Cloacibacillus*, *Gluconobacter*, *Oenococcus*, *Acetobacter*, *Komagataeibacter*, and *Liquorilactobacillus*. In the control group, they were *Pasteurella*, *Anaerobiospirillum*, *Atopobium*, *Mobiluncus*, *Bilophila*, *Fusobacterium*,

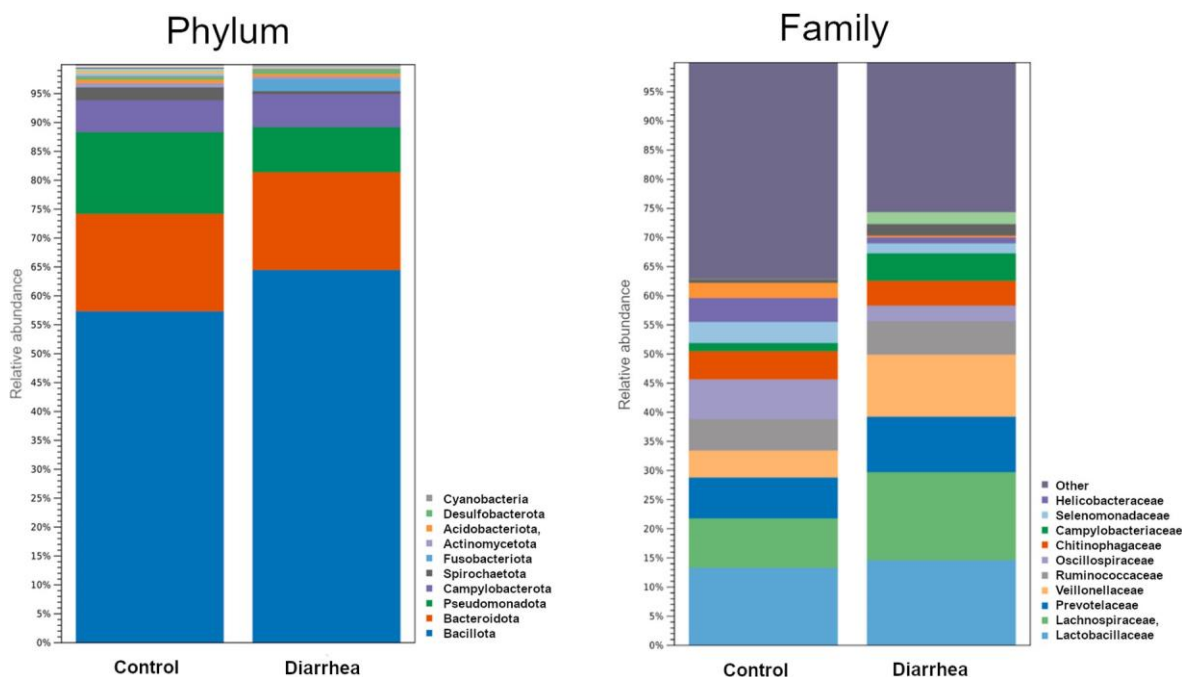


Fig. 1. Relative abundance of the main phyla and families in post-weaning pigs with and without diarrhea. The bar graph shows the 10 most abundant phyla and families in the control and diarrhea groups, highlighting the differences in microbial composition between the conditions. The phyla *Bacillota*, *Bacteroidota* and *Pseudomonadota* together represented almost 90 % of the total relative abundance.

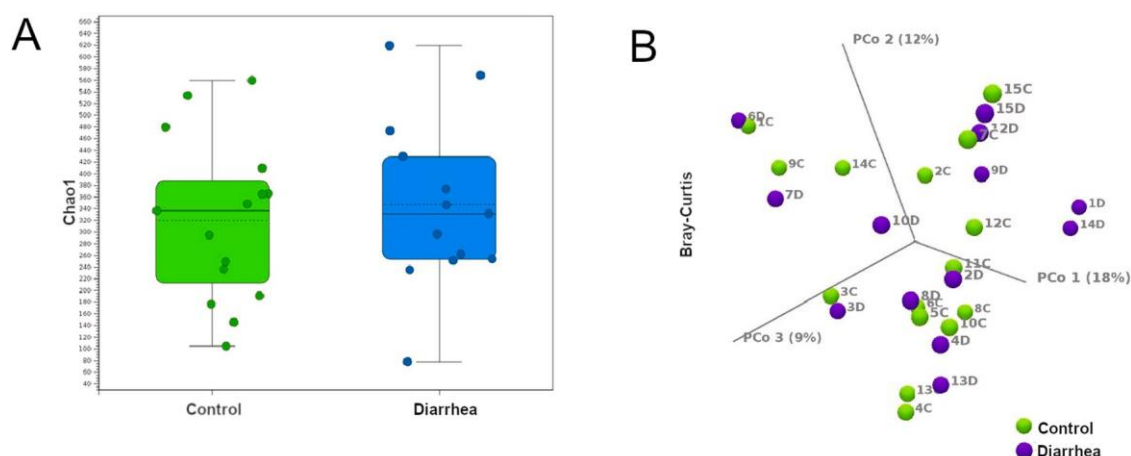


Fig. 2. Alpha and Beta diversities analyses of the intestinal microbiota in pigs with and without diarrhea. A. Alpha diversity was assessed by the Chao1 index, and comparison between groups was performed using the Mann-Whitney test, with no observed statistically significant differences ($p > 0.05$). B. Principal Coordinates Analysis (PCoA) based on the Bray-Curtis distance was used to assess beta diversity, and the PERMANOVA test indicated no significant differences between groups ($p > 0.05$).

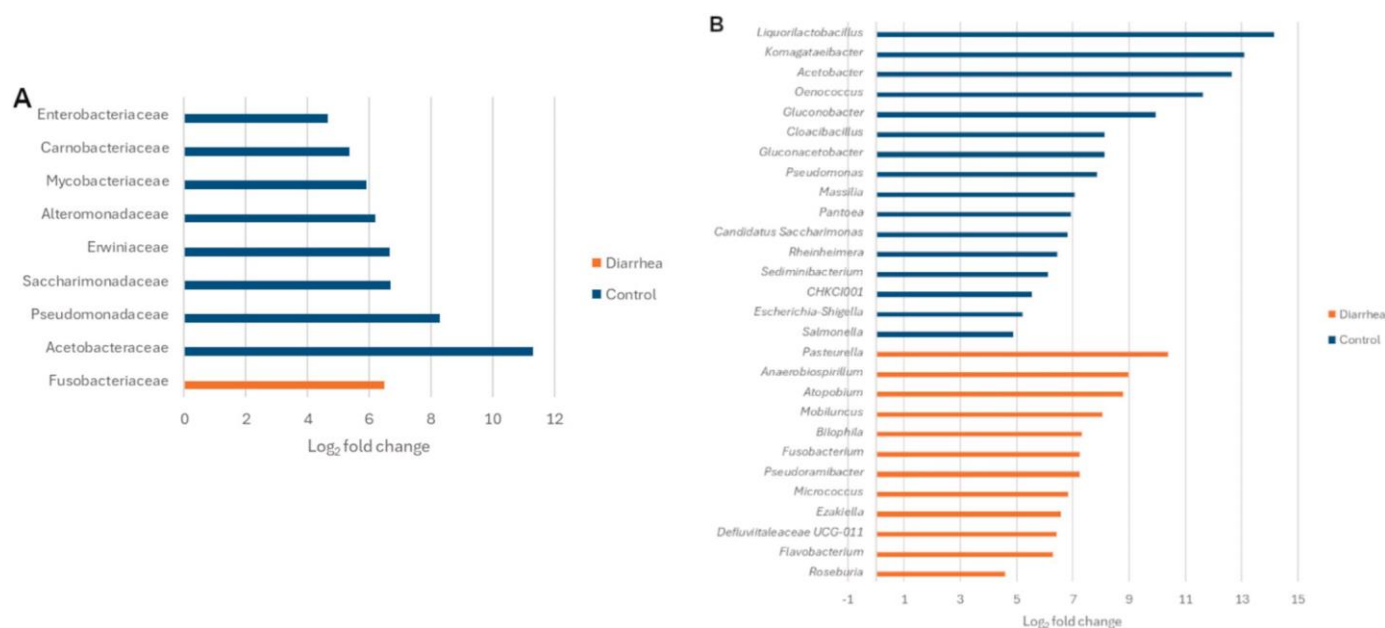


Fig. 3. Differential abundances expressed as log₂ fold-change at the taxonomic levels of (A) Family and (B) Genus. The taxa marked in orange are enriched in animals with diarrhea, whereas those in blue in healthy animals. A GLM was used to analyze differences in taxa abundance with a Likelihood Ratio test for group-wide comparisons. Only taxa with statistical significance ($p < 0.05$) are shown.

Pseudoramibacter, *Micrococcus*, *Ezakiella*, *Defluviitaleaceae*, *UCG-011*, *Flavobacterium*, *Roseburia*, *Salmonella*, *Escherichia-Shigella*, *CHKC1001*, and *Sediminibacterium* (Fig. 3B; Supplementary Table 3).

3.5. Metabolic functional prediction

In the analyzed groups, a total of 13 metabolic pathways differentially present in the analyzed groups were identified. In the diarrhea group, three pathways related to the degradation of aromatic compounds (2-aminophenol, 4-chloronitrobenzene, and 4-nitrotoluene II), novobiocin biosynthesis (secondary metabolite biosynthesis), mannosyl-fructose biosynthesis (carbohydrate biosynthesis), and energetic metabolism (reductive tricarboxylic acid cycle) were identified to be overrepresented when compared to control group. On the other hand, in the control group, five pathways related to amino acid degradation (trans-3-hydroxy-L-proline, L-lysine, L-threonine I, and two

superpathways, L-threonine and L-lysine), carbohydrate degradation (trehalose III), and fermentation ((S)-propano-1,2-diol) were enriched (Fig. 4).

4. Discussion

Characterizing the intestinal microbiota of suckling piglets is important for production, as it can help to predict the occurrence and severity of post-weaning diarrhea and influence growth performance in adulthood [19]. In this study, taxonomic analysis of gut microbiota sequences revealed a predominance of *Bacillota*, followed by *Bacteroidota* and *Pseudomonadota* in both groups. These phyla have been highlighted as the major constituents of swine intestine, regardless of age [20]. Other commonly present phyla include *Actinomycetota* and *Spirochaetes* [6,21]. While the overall taxonomic profile remains stable, microbial shifts below the phylum level can occur in response to intestinal

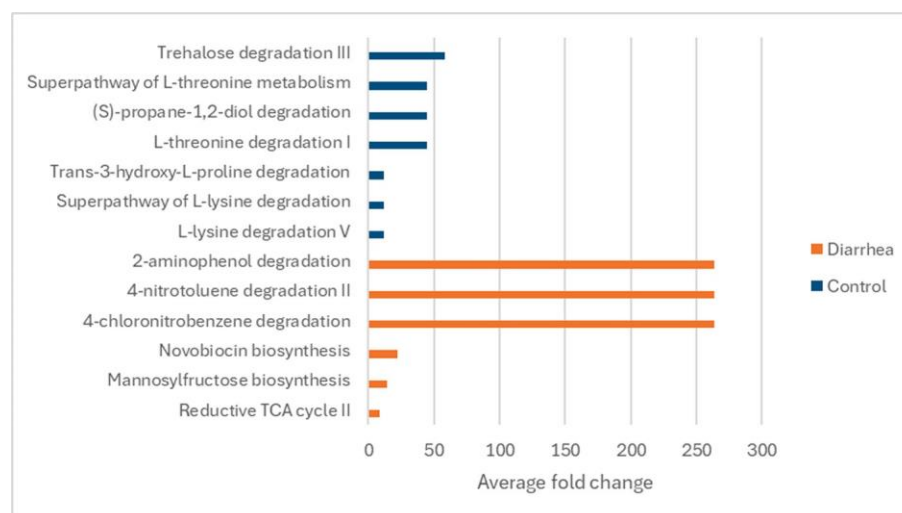


Fig. 4. Prediction of functional microbial metabolic pathways (PICRUSt) using the MetaCyc and Enzyme Commission (EC) databases. Pathways in blue are enriched in the control group and pathways in orange are enriched in the diarrhea group. Differential abundances expressed as $\log_2$ fold-change. A GLM was used to analyze differences in pathways abundance with a Likelihood Ratio test for group-wide comparisons. Only taxa with statistical significance ($p < 0.05$) are shown.

disturbances [21].

The most frequent families in our research were *Lactobacillaceae*, *Lachnospiraceae*, *Prevotellaceae*, *Veillonellaceae*, *Ruminococcaceae*, and *Oscillospiraceae* in both groups. Some, like *Prevotellaceae*, include bacteria that metabolize plant polysaccharides and glycans, likely adapting to dietary changes in weaned piglets [22]. Others, such as *Veillonellaceae* or *Oscillospiraceae*, can utilize minor compounds like plant monosaccharides (e. g. tagatose) or microbial fermentation byproducts like lactate, converting them into short-chain fatty acids [1]. The dominance of *Lactobacillaceae*, *Lachnospiraceae*, and *Ruminococcaceae* during the post-weaning phase has also been reported in previous studies, which observed that stress associated with the weaning transition resulted in episodes of diarrhea and reduced growth [23].

Regarding the bacterial communities, no significant differences in alpha and beta diversity metrics were observed between the groups, meaning that no overall diversity shifts occurred in the evaluated animals. The exclusive use of rectal swabs for sample collection limited the identification of the specific origin of diarrhea. Additionally, affected animals were separated at symptoms onset, introducing potential bias due to variability in the timing of diarrhea manifestation. It is important to point out that all animals, regardless of group, were from the same breed, were at the same growth phase, received the same feed and followed the same vaccination and antibiotic therapy [24] protocols on the farm, which may have influenced microbial diversity. Although this is a limitation, it also allows us to access the reality of a field situation.

Studies have shown that pigs receiving prophylactic antibiotics exhibit lower bacterial diversity, suggesting that these compounds alter the intestinal microbiota. Furthermore, antibiotic use in the pre- and postnatal period has been associated with reduced microbial diversity and an increased frequency of antibiotic-resistant bacteria, potentially leading to dysbiosis [25]. Research also indicates that antibiotics induce rapid shifts in gut microbiota composition, with changes detectable within two days in the ileum and within seven days in feces [26]. These factors may have contributed to the lack of differences in microbial diversity between the analyzed groups and the persistence of diarrhea.

The use of antibiotics in livestock for prophylaxis and growth promotion contributes to antimicrobial resistance and environmental contamination, posing risks to animal, human, and ecosystem health [27]. To address this, the WHO launched the Global Antimicrobial Resistance and Use Surveillance System (GLASS) in 2015, alongside initiatives from the OIE and FAO to combat resistant pathogens [69,70]. In Brazil, the Ministry of Agriculture and Livestock (MAPA)

implemented the second phase of the National Action Plan for Antimicrobial Resistance (PAN-BR AGRO) (2023–2027) to mitigate its impact on agriculture and public health [28]. In accordance with that, recent studies highlight alternatives like essential oils, prebiotics, probiotics, and zinc oxide as effective antibiotic replacements, promoting gut health and improving piglet performance [24,29,30]. These findings support reducing or eliminating antibiotics in pig farming as a viable strategy to curb antimicrobial resistance. Therefore, studies like ours can help in understanding the changes in the intestinal microbiota related to postweaning diarrhea and potentially provide new insights to produce alternatives to antibiotics.

Despite the absence of significant differences in alpha and beta diversities, differential abundance analysis revealed that one phylum and eight bacterial families were enriched in the control group, whereas only one family was enriched in the diarrhea group. Additionally, 12 genera were enriched in the diarrhea group and 16 in the control group. These findings suggest that dysbiosis may be present to some extent and could be associated with post-weaning diarrhea in pigs. Other studies have also reported no significant changes in alpha and beta diversity under diarrhea conditions in weaning pigs [31], suggesting that not always global microbiota alterations are observed in different health conditions, as it is often a stable structure. However, subtle shifts in microbial composition can still be detected. Notably, most of the enriched taxa were not among the top 10 most abundant in the evaluated microbiota. Research has shown that low-abundance microbial populations play a key role in pathogen-host interactions [67]. Even in small quantities, intestinal commensal bacteria contribute significantly to immune system development. These findings highlight their crucial role in immune maturation and modulation despite their low prevalence.

In the control group, the superphylum *Patescibacteria* was enriched. These ultra-small bacteria have a simplified genome, unusual ribosomal compositions, and sometimes intron sequences within *16S rRNA* genes, a rare feature in bacteria [32]. They are highly diverse and inhabit various ecosystems but have limited cellular synthesis capability [33]. Most cultured *Patescibacteria* live as obligate epibionts, adhering to and proliferating on other bacteria [36]. A Danish study identified *Patescibacteria* as one of the most abundant phyla in the colons of pigs, regardless of colonic inflammation [35]. Similarly, a Chinese study on pig microbiota reported *Patescibacteria* as highly abundant in healthy animals [34]. These findings align with our results, as both the Chinese study and ours observed a high prevalence of this phylum in healthy pigs. Although the Danish study also detected *Patescibacteria* in pigs with

colonic inflammation, the consistent association between this phylum and health in multiple studies supports its potential role in maintaining gut homeostasis.

In the diarrhea group, the *Fusobacteriaceae* family was enriched. *Fusobacteria* are known for their inflammatory potential, triggering host immune responses against pathogens. While inflammation is essential for homeostasis, infection defense, and tissue repair, uncontrolled inflammation can lead to cell damage and contribute to chronic diseases and cancer. *Fusobacterium*, particularly *Fusobacterium nucleatum*, is linked to colorectal cancer and inflammatory bowel disease in humans and is frequently detected in intestinal lesions and feces [37]. In piglets infected with porcine epidemic diarrhea virus (PEDV), increased *Fusobacterium* levels exacerbate intestinal inflammation [38]. This bacterium damages the epithelial barrier and worsens diarrhea by inducing pro-inflammatory cytokines, such as IL-1 α [39]. Luo et al. [40] reported a higher abundance of *Fusobacterium* in pigs during the weaning phase, noting that its presence can be up to four times greater in pigs with diarrhea than in healthy individuals. These findings support the association between *Fusobacterium* and intestinal inflammation, potentially explaining its presence in diarrheic piglets in our study.

The genus *Liquorilactobacillus*, enriched in the control group, has probiotic properties, including microbiota regulation, colitis relief, and strong antimicrobial activity [41]. Additionally, *Komagataeibacter*, *Acetobacter*, and *Gluconobacter* were enriched in this group. These genera produce bacterial cellulose by fermenting acetic acid and converting sugars like glucose, fructose, and sucrose [42,68]. The presence of these enriched genera in the control group suggests a potential protective role against diarrhea.

Other enriched genera in the control group were *Escherichia/Shigella*. These bacteria are naturally abundant in the piglet gut during the suckling phase, when the immune system is developing. While some species are opportunistic pathogens [43], maternal IgA antibodies from milk protect piglets against infections [19]. *Salmonella*, another pathogenic genus enriched in the control group, often causes asymptomatic infections in pigs, allowing prolonged fecal shedding [44]. As a major cause of foodborne illness worldwide, *Salmonella* from pork products is estimated to account for 15–23% of human salmonellosis cases in Europe [45]. Its ability to persist as a commensal in livestock complicates detection and control efforts [45,46].

The genus *Pseudomonas* was enriched in the control group. While primarily a respiratory pathogen, it has been detected in the gut of mice with dysbiosis, where *Bacteroidota* and *Bacillota* help to prevent *P. aeruginosa* colonization, whereas *Proteobacteria* promote it [47]. In pigs, respiratory infections caused by *Pseudomonas*, *Bordetella*, and *Burkholderia* can trigger gastrointestinal symptoms, like diarrhea, stimulating the immune system and damaging intestinal epithelial cells, increasing intestinal permeability and inflammation [48]. A rat model showed that *P. aeruginosa* can spread via the bloodstream to the gallbladder, multiply and migrate to the intestines and feces, facilitating transmission to other animals [49]. Its presence in the control group suggests potential transmission among animals, despite the absence of clinical symptoms at sampling.

The genus *Pasteurella* showed to be enriched in the diarrhea group. Although it is widely known for its association with respiratory infections, *Pasteurella* has also been described in weaned diarrheic pigs [50], which reinforces that dysbiosis associated with post-weaning diarrhea occurs in piglets.

Studies conducted in different regions, including those by Luo et al. [40], Kong et al. [50], Panah et al. [35], Wang et al. [51], Grandmont et al. [45] and our own, reveal consistent patterns in the gut microbiota of weaned pigs, both healthy and with diarrhea. These findings suggest that, despite geographic and management variations, universal factors—such as the transition to weaning and gut health play a key role in shaping microbiota composition and function, highlighting the presence of conserved biological mechanisms regulating gut microbial dynamics.

To further explore the microbiota's role in both groups, we

performed a functional prediction analysis, revealing a higher abundance of three aromatic compound degradation pathways (2-aminophenol, 4-chloronitrobenzene, and 4-nitrotoluene II) in the diarrhea group. According to PubChem (National Institutes of Health (NIH)), 2-aminophenol is a bacterial metabolite with an aerobic metabolic pathway described in *Pseudomonas*, *Bordetella*, and *Burkholderia* species [52]. *Bordetella*, a known user of this compound, is a causative agent of atrophic rhinitis in pigs, often co-occurring with *Pasteurella* in infected animals [53]. In our study, *Pasteurella* was the most enriched genus in the diarrhea group. Given that it is excreted in feces, its high abundance in these animals may reflect ongoing infection.

The degradation of the compounds 4-chloronitrobenzene and 4-nitrotoluene, enriched in the diarrhea group, with genes identified in *Pseudomonas* and *Mycobacterium* species [54]. These pathways allow bacteria to use these compounds as a source of nitrogen, carbon, and energy, releasing ammonia under aerobic conditions [55]. The metabolic versatility of opportunistic bacteria, such as *Pseudomonas aeruginosa*, contributes to their ability to colonize the gastrointestinal tract of hosts.

Amos et al. [56] identified antimicrobial-resistant *Pseudomonas* in pig feces, emphasizing its presence in the gastrointestinal tract and the need for responsible antimicrobial use in pig farming. Although *P. aeruginosa* is not directly linked to diarrhea in pigs, its role as an opportunistic bacterium suggests it may contribute to intestinal dysbiosis and exacerbate gastrointestinal disorders. Additionally, its ability to metabolize nitroaromatic compounds may enhance intestinal persistence and be associated with pathogenicity and colonization. This is particularly relevant in weaning pigs with immature immune systems, increasing their susceptibility to infections.

The reductive tricarboxylic acid (TCA) cycle was also enriched in the diarrhea group. According to the MetaCyc database, this pathway has both anabolic and catabolic components. The TCA cycle primarily converts fuel sources into energy, though several intermediates also serve as precursors for biosynthetic processes, such as the synthesis of sugars, lipids and amino acids [57]. The cycle is tightly regulated by physiological signals to optimize energy production and cellular function [58]. Stress from early weaning has been shown to reduce the expression of proteins related to the TCA cycle, β -oxidation and glycolysis in piglet enterocytes, particularly in the upper and middle jejunal villi [59]. This may explain the increased activity of the TCA reductase cycle in animals with diarrhea.

The metabolic pathway of trans-3-hydroxy-L-proline was enriched in the control group. This pathway is related to L-proline degradation [60]. Hydroxyproline (Hyp), an amino acid abundant in sow's milk, has antioxidant and anti-inflammatory properties. It is a precursor to glycine and is metabolized by intestinal microorganisms via the enzyme hydroxyproline dehydratase (HypD) [61]. Tang et al. [62] demonstrated that Hyp improves the intestinal health in weaned piglets by enhancing the intestinal barrier integrity and regulating microbiota. This explains the increased activity of this pathway in the control group, emphasizing its role in maintaining homeostasis and metabolic adaptation.

The control group also showed enrichment in two metabolic pathways for L-lysine degradation, which varies among microorganisms, animals, and plants. While many initial reactions are well-studied, some subsequent steps remain poorly understood. These pathways involve up to ten different routes for L-lysine degradation [54]. Lysine is an essential amino acid involved in hormone, enzyme, and antibody production, as well as calcium absorption, muscle formation, and recovery. It also plays a key role in treating viral infections, anxiety, migraines, mood disorders, Alzheimer's, alopecia, and certain cardiovascular diseases [63]. The enrichment of such pathways in control group indicates that it may be related to the health status of the gut in pigs.

Additionally, in the control group, L-threonine degradation pathways were enriched. L-threonine is an essential amino acid that supports aging prevention, immunity and disease resistance [64]. Deficiencies can lead to immune dysfunction and growth issues. L-threonine is widely

used in the food and pharmaceutical industries [64]. This amino acid is especially provided for piglets and young poultry, among other animals. In pig feeding, it is the second limiting amino acid, aiding growth, mucin production for improved digestion and nutrient absorption, and playing a key role in protein synthesis and energy metabolism [65].

Bacterial taxa and metabolic pathways associated with inflammatory and pathogenic microorganisms were identified in diarrheic pigs, reinforcing the link between dysbiosis and gastrointestinal disorders. Dysbiosis, characterized by reduced microbial diversity and the predominance of pro-inflammatory taxa (e.g., *Pseudomonadota*), leads to metabolic dysfunction, including decreased production of short-chain fatty acids (SCFAs) and increased inflammatory metabolites [29]. The transition from a liquid to a solid diet during weaning exacerbates these changes, compromising the intestinal barrier, reducing mucin production, and increasing epithelial [66]. Combined with the immaturity of the gastrointestinal tract and immune system, these factors heighten susceptibility to gastrointestinal diseases, emphasizing the need for strategies to modulate microbiota and mitigate the effects of dysbiosis [10].

This study has several limitations that should be considered. The small sample size may restrict the generalizability of the findings, and antibiotic use in both groups may have influenced the intestinal microbiota. Additionally, the lack of detailed animal records such as sex, diarrhea severity and exact onset, and the absence of pre-diarrhea microbiota data limit the ability to assess microbial shifts. Lastly, the absence of longitudinal follow-up prevents the analysis of microbiota dynamics over time. Future studies with larger sample sizes and continuous monitoring could provide a more comprehensive understanding of these effects.

5. Conclusions

The results indicate that although there is no significant difference in the overall bacterial composition in piglets with and without diarrhea, a shift in the microbiota and metabolic pathways was detected in terms of differential abundance. Most of the taxa enriched in the diarrhea group were associated with inflammatory microorganisms involved in pathogenic processes. Additionally, enriched metabolic pathways related to bacterial metabolism were observed, suggesting increased microbial activity that may be contributing to diarrhea condition.

In the control group, all enriched metabolic pathways are fundamental for pig development. Likewise, the enriched taxa identified in the control group provides valuable insights for developing alternatives to antibiotic use during the post-weaning phase. Implementing probiotics or other feed additives capable of positively modulating the intestinal microbiota could serve as a promising strategy to prevent diarrhea, promote gut health, and enhance the zootechnical performance of pigs. These findings are relevant for pig industry, especially in evaluated area, since environmental conditions and management practices can directly influence microbial composition.

CRediT authorship contribution statement

Karen Karine da Rosa Dias: Data curation, Writing – original draft, Methodology, Conceptualization, Software. **Caroline Tochetto:** Data curation, Writing – review & editing, Methodology. **Ana Paula Muterle Varela:** Writing – review & editing, Methodology. **Marcia Regina Loiko:** Investigation, Writing – review & editing, Conceptualization. **Anne Caroline dos Santos:** Conceptualization, Writing – review & editing, Investigation. **Jeverson Frazzon:** Writing – review & editing, Resources. **Paulo Michel Roehs:** Writing – review & editing, Supervision, Resources. **Fabiana Quos Mayer:** Writing – review & editing, Methodology, Resources, Supervision, Conceptualization.

Data availability

The raw *16s rRNA* sequence data has been deposited at the National Center for Biotechnology Information (NCBI) and is available under the BioProject accession number PRJNA1178803.

Funding

P.M.R. and F.Q.M. are CNPq research fellows (309024/2020-0 & 315255/2021-8). KKRD has a PhD scholarship from CAPES (88887.664183/2022-00).

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2025.107878>.

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Supplementary Table 1

Name	Number of reads	Avg.length	Trimmed sequences	Trimmed (broken pairs)	Total number of reads after trim	Percentage trimmed (%)	Avg.length after trim
1C (paired)	170.746	250,89	169288	726	170.014	99,57	239,77
1D (paired)	233.352	250,94	231368	985	232.353	99,57	240,3
2C (paired)	289.210	250,9	286938	1120	288.058	99,6	235,21
2D (paired)	249.294	250,95	247004	1135	248.139	99,54	239,07
3C (paired)	214.248	250,95	212446	889	213.335	99,57	237,86
3D (paired)	201.286	250,91	199426	919	200.345	99,53	236,01
4C (paired)	230.108	250,95	227888	1096	228.984	99,51	233,5
4D (paired)	239.500	250,98	237160	1162	238.322	99,51	236,01
5D (paired)	48.712	250,66	48128	291	48.419	99,4	234,12
5C (paired)	253.398	250,94	251402	990	252.392	99,6	237,41
6C (paired)	208.712	250,91	206842	929	207.771	99,55	236,84
6D (paired)	165.984	250,81	163966	990	164.956	99,38	230,48
7C (paired)	160.662	250,97	158856	894	159.750	99,43	233,33
7D (paired)	213.456	250,93	210914	1257	212.171	99,4	235,16
8C (paired)	223.014	250,91	221202	890	222.092	99,59	235,61
8D (paired)	168.810	250,95	166456	1159	167.615	99,29	231,67
9D (paired)	290.114	250,97	287054	1505	288.559	99,46	236,17
9C (paired)	166.964	250,89	165516	721	166.237	99,56	239,57
10C (paired)	215.780	250,96	213788	983	214.771	99,53	233,67
10D (paired)	234.124	250,96	231788	1148	232.936	99,49	237,07
11C (paired)	199.712	250,98	197698	995	198.693	99,49	232,9
11D (paired)	854	250,93	800	26	826	96,72	165,08
12C (paired)	234.022	250,98	231868	1064	232.932	99,53	230,87
12D (paired)	797.382	250,98	788798	4238	793.036	99,45	234,82
13C (paired)	259.720	250,9	257264	1217	258.481	99,52	235,59
13D (paired)	212.042	250,95	210276	865	211.141	99,58	236,15
14C (paired)	169.166	250,91	167372	892	168.264	99,47	240,1
14D (paired)	239.062	250,96	237182	929	238.111	99,6	241,08
15C (paired)	192.054	250,96	189908	1060	190.968	99,43	234,2
15D (paired)	200.890	250,96	199066	904	199.970	99,54	235,99

Supplementary Table 2

A tabela completa, em função de seu extenso formato, está disponível para consulta em:

<https://www.sciencedirect.com/science/article/pii/S0882401025006035>

Supplementary Table 3 - Phylum

Name	Due to Type - Max group mean	Due to Type - Log ₂ fold change	Due to Type - Fold change	Due to Type - P- value	Due to Type - FDR p-value	Due to Type - Bonferroni
Acidobacteriota	504,8667	-0,152301	-1,111341	0,821270	0,858154	1
Actinobacteriota	552,0000	-0,813593	-1,757584	0,103509	0,277589	1
Ascomycota	0,2308	0,549717	1,463799	0,633917	0,771726	1
Bacteroidota	17334,3077	-0,977872	-1,969557	0,051538	0,206153	1
Basidiomycota	0,2667	-2,072471	-4,206064	0,133225	0,277589	1
Bdellovibrionota	80,4000	-1,563472	-2,955642	0,134896	0,277589	1
Campylobacterota	5909,0000	0,183320	1,135494	0,858154	0,858154	1
Cyanobacteria	252,6923	0,271118	1,206742	0,683949	0,797941	1
Deferribacterota	296,4000	-1,562696	-2,954053	0,270683	0,398902	1
Deinococcota	0,2308	1,776108	3,425010	0,163930	0,286877	1
Desulfobacterota	642,5385	0,972813	1,962664	0,409595	0,546126	1
Elusimicrobiota	0,3077	2,022357	4,062470	0,081956	0,277589	1
Euryarchaeota	349,4000	-3,773695	-13,677141	0,002091	0,021827	0,058540
Fibrobacterota	6,2308	3,686483	12,874847	0,006224	0,043566	0,174263
Firmicutes	66245,8462	-0,452108	-1,368038	0,534035	0,679681	1
Fusobacteriota	2081,3077	6,226761	74,893094	0,002339	0,021827	0,065481
Myxococcota	145,3333	-0,230241	-1,173031	0,818455	0,858154	1
Parabasal	1,9333	-1,782682	-3,440651	0,146027	0,277589	1
Patescibacteria	46,9333	-7,741498	-214,004632	0,000026	0,000721	0,000721
Planctomycetota	207,5333	-0,944996	-1,925183	0,148709	0,277589	1
Proteobacteria	11623,5333	-0,968530	-1,956845	0,096784	0,277589	1
Spirochaetota	1804,3333	-1,880941	-3,683151	0,114373	0,277589	1
Sumerlaeota	0,2000	-1,550924	-2,930048	0,236102	0,367270	1
Synergistota	50,2000	-3,069912	-8,397223	0,014320	0,066827	0,400960
Thermoplasmatota	47,6667	-0,999204	-1,998896	0,407371	0,546126	1
Verrucomicrobiota	173,0769	0,257732	1,195598	0,834034	0,858154	1
WPS-2	18,7692	1,754895	3,375019	0,186645	0,307415	1

Supplementary Table 3 - Family

Name	Diarrhea vs. Control - Max group mean	Diarrhea vs. Control - Log ₂ fold change	Diarrhea vs. Control - Fold change	Diarrhea vs. Control - P-value	Diarrhea vs. Control - FDR p-value	Diarrhea vs. Control - Bonferroni
Acetobacteraceae	1927,13333	-11,31865	-2554,18480	0,000000	0,00000000	0,00000000
Alteromonadaceae	2,46667	-6,21068	-74,06318	0,000103	0,00258260	0,02066080
Carnobacteriaceae	3,40000	-5,36994	-41,35356	0,000192	0,00398160	0,03843804
Enterobacteriaceae	1484,06667	-4,67005	-25,45811	0,000045	0,00129906	0,00909340
Erwiniaceae	3,53333	-6,67276	-102,02398	0,000036	0,00120165	0,00720989
Fusobacteriaceae	2081,30769	6,50060	90,54760	0,000012	0,00048886	0,00244432
Mycobacteriaceae	4,73333	-5,90970	-60,11685	0,000199	0,00398160	0,03981605
Pseudomonadaceae	2015,40000	-8,27778	-310,35529	0,000000	0,00000000	0,00000000
Saccharimonadaceae	46,93333	-6,69331	-103,48720	0,000001	0,00004009	0,00012026

Supplementary Table 3 - Genus

Name	Diarrhea vs. Control - Max group mean	Diarrhea vs. Control - Log ₂ fold change	Diarrhea vs. Control - Fold change	Diarrhea vs. Control - P-value	Diarrhea vs. Control - FDR p-value	Diarrhea vs. Control - Bonferroni
Acetobacter	364,333333	-12,648254	-6419,538047	2,51163E-14	3,37814E-12	1,35126E-11
Anaerobiospirillum	123,538462	8,981196	505,370022	0,000000	0,000000	0,000002
Atopobium	35,923077	8,797114	444,831217	0,000000	0,000016	0,000198
Bilophila	12,769231	7,329679	160,861942	0,000011	0,000234	0,006084
Candidatus Saccharimonas	46,933333	-6,804569	-111,783940	0,000001	0,000035	0,000556
CHKCI001	36,400000	-5,561034	-47,210429	0,000041	0,000768	0,022262
Cloacibacillus	24,000000	-8,131360	-280,403335	0,000001	0,000028	0,000391
Defluviitaleaceae UCG-011	14,384615	6,426621	86,021267	0,000052	0,000899	0,027871
Escherichia-Shigella	602,466667	-5,216853	-37,190252	0,000002	0,000051	0,000872
Ezakiella	14,076923	6,582816	95,857262	0,000009	0,000203	0,005077
Flavobacterium	6,461538	6,286146	78,040227	0,000092	0,001419	0,049682
Fusobacterium	2079,384615	7,256285	152,883091	0,000000	0,000016	0,000211
Gluconacetobacter	7,400000	-8,128098	-279,770158	0,000003	0,000074	0,001386
Gluconobacter	89,200000	-9,957391	-994,198819	0,000000	0,000000	0,000000
Komagataeibacter	1411,400000	-13,092636	-8735,263579	1,24947E-18	3,36108E-16	6,72216E-16
Liquorilactobacillus	4147,266667	-14,148631	-18161,949777	1,35852E-21	7,30885E-19	7,30885E-19
Massilia	29,600000	-7,073740	-134,712486	0,000000	0,000004	0,000040
Micrococcus	9,076923	6,846492	115,079917	0,000047	0,000843	0,025289
Mobiluncus	21,307692	8,055534	266,046301	0,000003	0,000086	0,001712
Oenococcus	478,200000	-11,628967	-3167,144762	1,2679E-14	2,27377E-12	6,8213E-12
Pantoea	3,533333	-6,949884	-123,629946	0,000022	0,000428	0,011991
Pasteurella	734,461538	10,384357	1336,604264	6,9478E-14	7,47583E-12	3,73792E-11
Pseudomonas	2015,266667	-7,868695	-233,729288	0,000000	0,000000	0,000001
Pseudoramibacter	40,384615	7,239290	151,092636	0,000001	0,000030	0,000455
Rheinheimera	2,466667	-6,452705	-87,590678	0,000057	0,000965	0,030882
Roseburia	3025,230769	4,588025	24,051004	0,000006	0,000142	0,003263
Salmonella	873,400000	-4,888508	-29,620179	0,000005	0,000127	0,002787
Sediminibacterium	54,666667	-6,117662	-69,438409	0,000007	0,000146	0,003511

1. ARTIGO CIENTÍFICO 2

O segundo artigo desta tese teve por objetivo caracterizar e comparar o metagenoma viral de leitões desmamados com e sem diarreia pós-desmame. O artigo será submetido à revista *Veterinary Microbiology*.

Compositional Shifts in the Intestinal Virome of Weaned Piglets with Diarrhea

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Abstract

Post-weaning diarrhea in piglets represents a significant challenge in pig farming, affecting animal health and economic viability. Although the bacterial microbiome has been studied, the role of the intestinal virome in the pathogenesis of diarrhea remains poorly understood. This study characterized the gut virome of weaned piglets with and without diarrhea aiming to identify viral signatures associated with this condition. There were no significant differences in alpha diversity between groups, but beta diversity analysis revealed significant qualitative changes in viral composition. The group with diarrhea showed an increase in the relative abundance of *Bocaparvovirus* and *Rotavirus A* in differential abundance, both pathogenic eukaryotic viruses, and enrichment of bacteriophages that infect commensal or opportunistic bacteria (*Junavirus*, *Paclarkvirus*), accompanied by a reduction in the relative abundance of regulatory bacteriophages (*Phixviricota/Microviridae*). In contrast, healthy piglets had a viral community dominated by regulatory bacteriophages (*Peduoviridae*, *Phitrevirus phi3*) and commensal CRESS-DNA viruses (*Porprismacovirus*). Free-living amoeba viruses were detected exclusively in the control group, suggesting differentiated environmental exposure. It can be concluded that there is a reconfiguration of the intestinal virome in post-weaning diarrhea, characterized by the preservation of overall viral diversity, but with compositional changes where regulatory bacteriophages and commensal viruses are replaced by a viral signature of pathogenicity and dysbiosis. The selective presence of different viruses among the groups reinforces that components of the intestinal virome may reflect different states of equilibrium in the intestinal ecosystem and the influence of external factors.

Keywords: Viral metagenomics; gastrointestinal virome; swine virus; diarrhea; stress weaning.

Introduction

Pig farming is a segment of significant importance in the global animal protein market, pork being the second most consumed meat in the world (ABPA, 2024). Recent projections by the OECD-FAO (2024) indicate an expansion trend of around 8% in global consumption by 2033, corroborating its significant participation in the international agro-industrial complex. Brazil has a prominent position in the production and trade scenario and is the world's fourth largest producer and exporter (ABPA, 2024b). Modern pig farming adopts early weaning between two and four weeks of age, significantly earlier than the natural period that would occur between 12 and 18 weeks in the wild. This practice, although economically advantageous, presents considerable challenges due to the physiological immaturity of piglets, slowing down digestion and absorption (Qin et al., 2024). The still-developing digestive and immune systems, combined with the abrupt interruption of the supply of immunoglobulins and other bioactive components of mother's milk, result in increased susceptibility to pathogens and intestinal disorders (St-Pierre et al., 2023).

Early weaning exacerbates stress in piglets, compromising the function of the intestinal epithelial barrier and significantly increasing the risk of enteric dysfunctions. It also causes electrolyte imbalances in the gastrointestinal tract, characterized by a decrease in the expression of fluid absorption proteins and an increase in the expression of secretion proteins (Tang et al., 2022). These changes lead to the accumulation of fluid in the intestinal lumen and large intestine, often resulting in diarrhea that can progress to severe cases with dehydration and mortality (Tao et al., 2022).

The gut microbiome forms a complex ecosystem where various microorganisms interact and influence gut health. In a previous study conducted by our group, we observed that the main change in the bacterial microbiota of animals with post-weaning diarrhea lies in its composition (differential abundances), with no significant changes in other general parameters of the bacterial community when compared to healthy animals (Dias et al., 2025). This result highlighted the importance of investigating other components of the microbiome, especially the virome, as possibly altered in this condition. Other studies have already demonstrated that viruses have the potential to

modulate both the composition and functionality of the bacterial microbiota through complex interactions with host microorganisms, and can influence digestive, immunological and inflammatory processes, significantly impacting the intestinal health of piglets during the critical post-weaning period (Wang et al., 2023).

Although the role of viruses in the gut microbiome is not yet fully understood, recent studies suggest that these agents can contribute to maintaining microbial balance and strengthening the immune system (Zeng et al., 2024). Bacteriophages stand out in this context because they significantly influence the composition and functions of the bacterial microbiota. In addition, these viruses act as modulators of the immune response, participating in the regulation of innate, humoral and cell-mediated immunity (Leonard & Toro, 2023).

Considering the challenges faced during the weaning phase in pig farming, this study aimed to characterize the intestinal viral communities of piglets with and without diarrhea on a farm in southern Brazil. We sought to describe the viral diversity of these animals and investigate the relationship between the virome and intestinal health, to identify possible correlations with enteric disorders.

Materials and methods

Ethical approval

This study was approved by the UFRGS Animal Use Ethics Committee (CEUA, protocol 33558/2018), strictly following animal welfare standards to minimize animal discomfort and suffering.

Experimental design

The study was conducted with 30 weaned Duroc piglets, aged 33 days old in the nursery phase, from a certified swine breeding farm (CSBF) in Rio Grande do Sul, Brazil. Convenience sampling included 15 animals with diarrhea, characterized by the presence of liquid feces, and 15 healthy animals as a control group, randomly selected from those without clinical signs of diarrhea. The diarrheic piglets were kept together in the same pen, while the healthy animals were distributed in groups of 10 individuals per

pen. The sample size was determined by the availability of animals with symptoms at the time of collection, maintaining numerical parity between the groups to ensure comparability in the analysis.

Food and health management

The nutritional and health protocol for the animals was implemented at two critical stages of development. During the dietary transition phase, at 15 days of age, the piglets received a pre-starter diet formulated with highly digestible ingredients, including fish meal, milk powder, soy, and corn, supplemented with a vitamin-mineral complex, probiotics, prebiotics, and antioxidants to strengthen the immune system and promote intestinal health.

During the weaning period, established at 28 days of age, a comprehensive vaccination schedule was implemented, with the application of three immunobiologicals: autogenous vaccine against *Actinobacillus pleuropneumoniae* (Microvet, Brazil), M+PAC for *Mycoplasma hyopneumoniae* (MSD Animal Health, USA), and Circumvent PCV against *porcine circovirus type 2* (MSD Animal Health, USA). In addition, a preventive therapeutic regimen was adopted that included the administration of Norfloxacin 14 mg/kg (Farmaflox®, Farmabase™, Brazil) in drinking water between 21 and 28 days of age, as well as a single intramuscular dose of Tulathromycin 2.5 mg/kg (Draxin®, Zoetis™, USA).

Sample collection and storage

Rectal swabs were collected from all the animals (n = 15 per group) and placed in sterile tubes containing 400 µL of RNALater® (Thermo Fisher Scientific, USA). The samples were transported on ice on the same day as collection and stored at -70 °C until processing in the laboratory.

Sample processing and Viral nucleic acid extraction

The swabs with RNALater® were thawed and 800 µL of phosphate-buffered saline (PBS) were added. The samples were homogenized in sterile tubes containing sterile ceramic beads using a shaking device (Precellys® 24 - Bertin Technologies S.A.S,

Versailles, France), three times at 5,500 rpm for 20 seconds, with 30 second intervals between each cycle. For the extraction of viral nucleic acids, the samples were individually filtered with sterile 0.45 µm filters and centrifuged at 130,000 x g for 2 h at 4 °C on 25% sucrose cushions. Prior to filtration and centrifugation, sample volumes were adjusted with PBS to at least 5 mL. The supernatant was discarded and the resulting pellet was resuspended in 200 µL of ultrapure water and treated with endonucleases by adding 25 µL of DNase buffer (Thermo Scientific), 5 µL of RNase (Thermo Scientific) and 2 µL of turbo DNase (Thermo Scientific). The samples were incubated for 2 hours at 37 °C. Enzyme inactivation was performed by adding EDTA to a final concentration of 15 mM, followed by heating at 75°C for 10 minutes. The nucleic acids were subsequently extracted with MagMAX Pathogen DNA/RNA purification kit (Life Technologies, Inc.), according to the manufacturer's instructions.

Nucleic acid enrichment e virome sequencing

Viral DNA was enriched by multiple displacement amplification (MDA) using φ29 DNA polymerase, as described by Dean et al. (2002). Viral RNA was enriched using REPLI-g WTA Single Cell Kit (Qiagen), following the manufacturer's instructions. The enriched nucleic acids were quantified using the Qubit™ dsDNA HS Assay Kit on a Qubit 2.0 Fluometer (Invitrogen, Life technologies) and subsequently combined in equimolar amounts. The mixture was purified using the Agencourt AMPure XP (Beckman Coulter), according to the manufacturer's protocol. Library preparation was performed using the Nextera XT DNA Library Prep kit (Illumina), and sequencing was carried out on the Illumina Miseq platform using the Miseq Reagent kit v2 500 cycles (Illumina), generating 2 x 250 bp.

Bioinformatics analysis

The quality of the raw sequences was assessed by FastQC (phred score <20) and filtered with Trimmomatic to remove adapters, low-quality regions and short reads (<50 bp), with final re-evaluation by FastQC to ensure reliable data. For viral identification, the reads were aligned to the Viral RefSeq database using DIAMOND in blastx mode, with optimized parameters (E-value ≤1e-5, best hit per sequence and 40 threads). The

tabular result included alignment metrics and taxonomic IDs, allowing efficient viral classification without additional annotation steps. This approach ensured a sensitive and specific analysis, even for large volumes of data.

Relative abundance

The data normality and homogeneity of variances were assessed by Shapiro-Wilk and Levene tests. The relative abundance of viruses among the experimental groups was compared using the Mann-Whitney U test. To characterize the dominant viral profile, a descending ranking of the 10 most abundant taxa at each taxonomic level (Family, Genus, and Species) was performed, allowing the identification of the main viral representatives in both experimental conditions.

Alpha and beta diversities

The alpha diversity indices (viral richness, Shannon, Simpson, and Pielou) were calculated using the vegan package. All indices were previously subjected to normality (Shapiro-Wilk) and homoscedasticity (Levene) tests. Based on this assessment, the viral richness and Pielou uniformity indices, which met the parametric criteria, were analyzed using Student's t-test, while the Shannon and Simpson indices, which did not have normal distribution, were evaluated using the Mann-Whitney U test.

For beta diversity, two dissimilarity metrics were applied: Bray-Curtis (relative abundance) and Jaccard (presence/absence). The method comprised the calculation of distance matrices, followed by principal coordinate analysis (PCoA) to visualize dissimilarity patterns. The statistical significance of differences between groups was determined using the PERMANOVA test with 9,999 permutations, with graphical representation limited to PCoA plots, which included confidence ellipses and statistical annotations.

Differential abundance

Differential abundance analysis was performed using the DESeq2 package. The standardized workflow included: preprocessing with low-count filtering (<10 reads), normalization by size factors, dispersion estimation, and Wald testing. Strict significance

criteria ($p\text{-value} < 0.05$ and $\text{FDR} < 0.1$) were adopted to identify differentially abundant rates at multiple taxonomic levels (Family, Genus, Species). The results were visualized in horizontal bar charts and exported in appropriate formats to ensure transparency and analytical reproducibility.

Statistical analysis

Statistical analyses were performed in the R environment (R Core Team, 2025), using an integrated approach that included analyses of relative abundance (Mann-Whitney U test), alpha diversity (Student's t-test and Mann-Whitney U test), beta diversity (PERMANOVA with four dissimilarity metrics), and differential abundance (DESeq2). Data manipulation used the dplyr, tidyr, and stringr packages; graphical visualizations were developed with ggplot2 and ggforce; ecological analyses employed vegan; and phylogenetic analyses used phyloseq and ape. The level of significance was set at $p \leq 0.05$.

Results

Relative abundance

Metagenomic sequencing of the 30 samples (15 Control and 15 Diarrhea) generated 4,616,790 raw reads. After quality filtering, 49.83% of the sequences (2,300,618 reads) were retained for analysis, of which 11.59% (266,653 reads) were identified as viral. The Diarrhea group showed a higher average proportion of viral reads (13.95%) compared to the Control group (10.22%), with samples D2 (32.86%) and D7 (32.77%) exhibiting the highest viral loads (Supplementary Table S1). Taxonomic analysis revealed extensive viral diversity, comprising 14 phyla, 23 classes, 42 orders, 114 families, 1,185 genera, and 2,850 viral species.

At the *Family* level, the Control group showed dominance of *Microviridae* (55.89%) and *Smacoviridae* (20.69%), followed by *Parvoviridae* (6.81%) and *Genomoviridae* (2.04%). The Diarrhea group exhibited *Parvoviridae* (31.62%), *Microviridae* (30.81%) and *Smacoviridae* (27.48%) as the most representative families (Figure 1A).

Genus-level analysis revealed that the Control group was dominated by

Porprismacovirus (31.18%) and *Bocaparvovirus* (10.58%), with secondary contributions from *Astrithrvirus* (5.10%) and *Oengusvirus* (3.47%). In the Diarrhea group, *Bocaparvovirus* was the most abundant genus (40.24%), followed by *Porprismacovirus* (28.97%) (Figure 1B).

At species-level, the Control group had greater abundance of species like *Lepus americanus faeces* (6.59%) and *Lynx canadensis associated* (5.72%), while the Diarrhea group was characterized by *Bocaparvovirus ungulate 5* (12.23%) and *Porcine associated porprismacovirus* (12.14%) (Figure 1C).

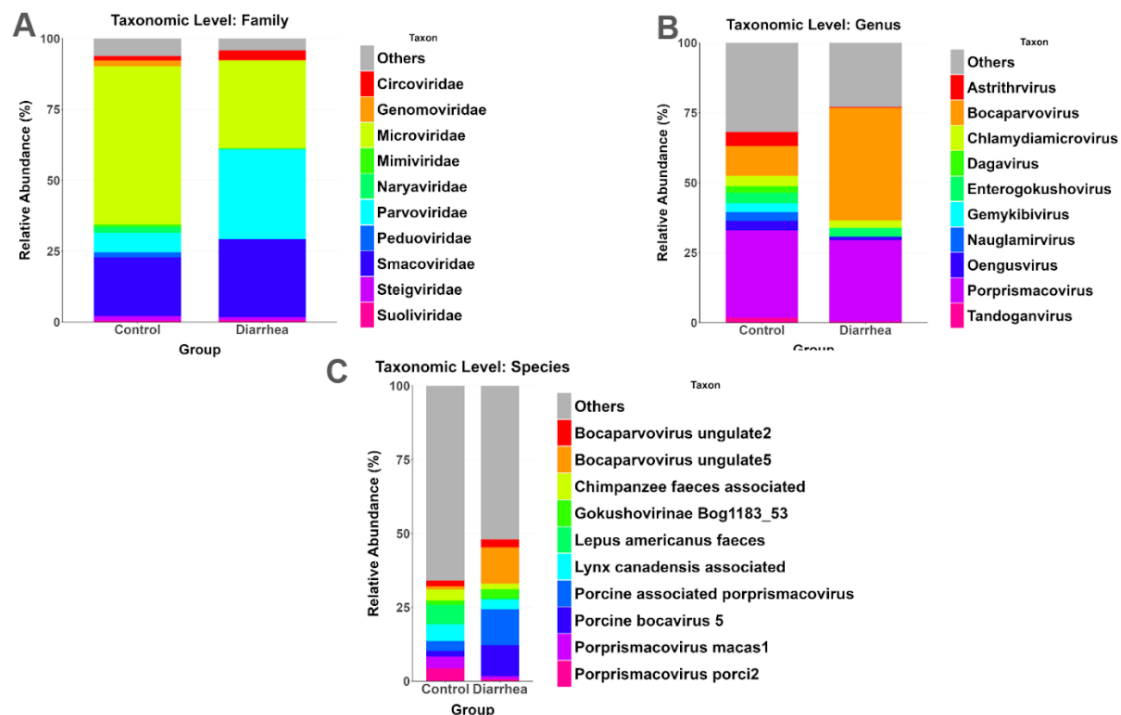


Figure 1. Relative viral abundance in the Control and Diarrhea groups. The stacked bars represent the proportional distribution of the main viral taxa. The taxonomic levels analyzed were: (A) Family, (B) Genus, and (C) Species.

Alpha and beta diversities

Viral richness ($t = -1,834$; $p = 0.077$), diversity (Shannon $U = 80$; $p = 0,187$; Simpson $U = 92$; $p = 0,412$), and evenness (Pielou $t = -0,345$; $p = 0.733$) indicate that the overall structure of viral communities remained stable regardless of clinical condition (Figure 2).

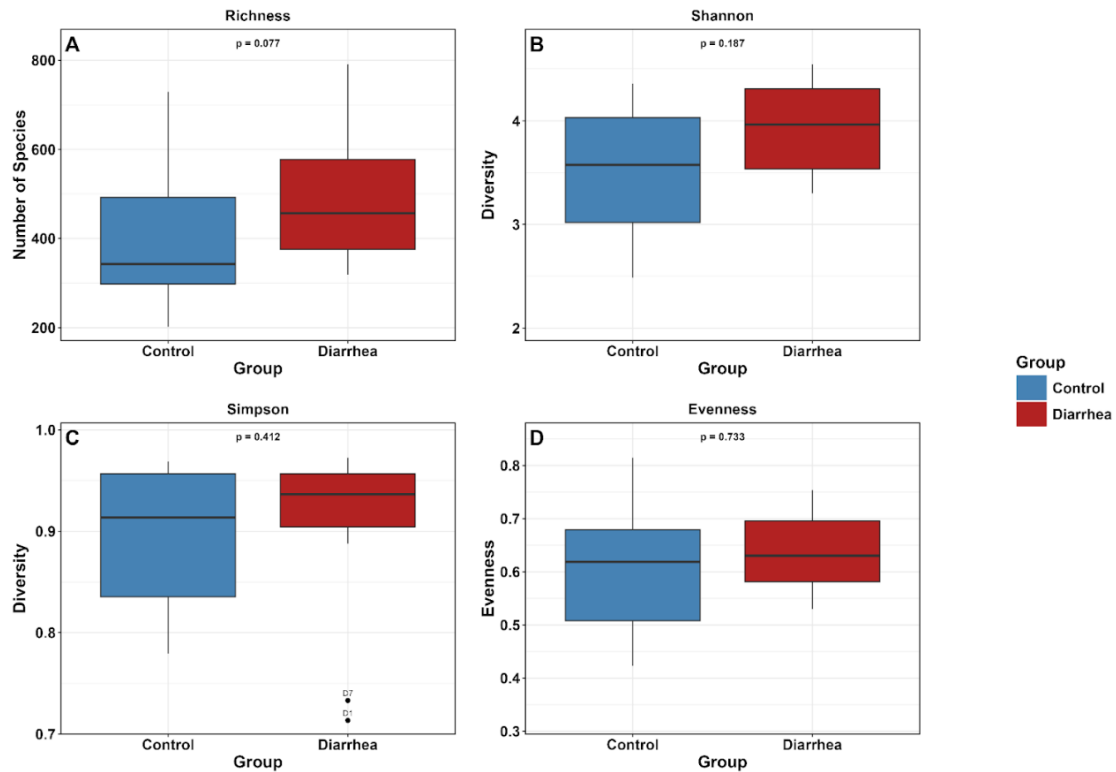


Figure 2. Alpha diversity metrics of the gut virome in Control and Diarrhea groups. (A) Species richness, representing the number of distinct viral species detected per sample. (B) Shannon diversity index, combining species richness and evenness. (C) Simpson diversity index, reflecting species dominance in the community. (D) Pielou's evenness, measuring the uniformity of species abundance distribution. All metrics show no statistically significant differences between groups ($p > 0.05$), indicating preserved overall viral diversity despite diarrheal condition, though individual variations are observed within groups.

For beta-diversity analysis, Bray-Curtis dissimilarity, which considers differences in relative virus abundance, revealed no significant separation between groups ($F = 1.382$; $R^2 = 0.047$; $p = 0.064$). In contrast, Jaccard dissimilarity, based exclusively on the presence and absence of viruses, showed a significant difference in viral composition between groups ($F = 1.549$; $R^2 = 0.052$; $p = 0.001$). These results indicate that, although the groups have similar profiles of relative viral abundance, they differ significantly in the presence/absence of specific viruses (Figure 3).

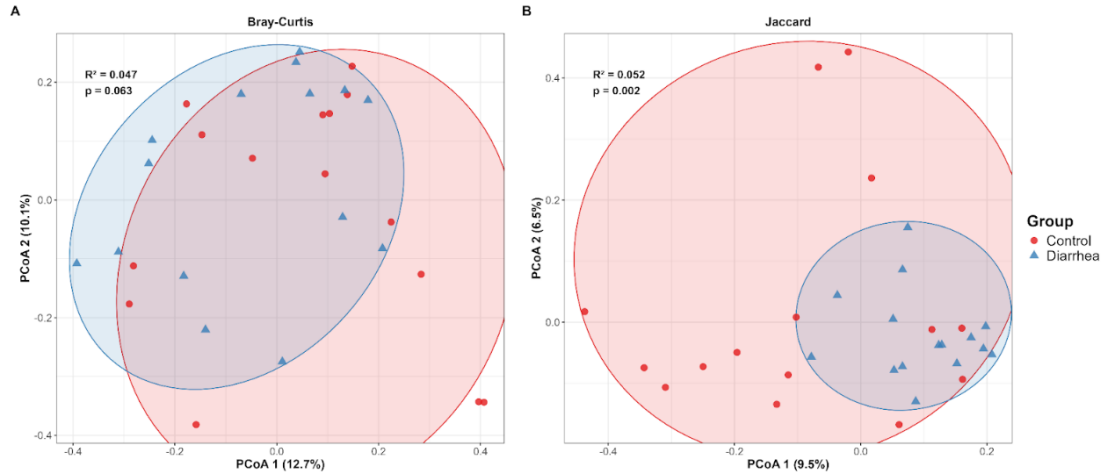


Figure 3. Beta-diversity showing principal coordinate analysis (PCoA) between microbial communities of control groups (red circles, $n=15$) and diarrhea (blue triangles, $n=15$). A) Bray-Curtis dissimilarity (relative abundance) reveals a non-significant separation ($R^2=0.047$; $p=0.064$). B) Jaccard dissimilarity (presence/absence) shows a significant difference in composition ($R^2=0.052$; $p=0.001$). Ellipses represent 95% confidence intervals. Percentages on the axes indicate variance explained by each principal component.

Differential abundance

At the family level, only two taxa showed significant differential signatures: *Genomoviridae* ($\log_2FC = +2.54$; $padj = 0.042$) was enriched in the Diarrhea group, while *Peduvoviridae* ($\log_2FC = -1.59$; $padj = 0.074$) was enriched in the Control group (Figure 4A). At the genus level, eleven significant taxa were identified, notably *Rotavirus* ($\log_2FC = +4.31$; $padj = 0.003$) and *Samwavirus* ($\log_2FC = +4.20$; $padj = 0.0004$) as enriched in Diarrhea, while *Dagavirus* ($\log_2FC = -2.83$; $padj = 0.00003$) and *Mieseafarmvirus* ($\log_2FC = -3.09$; $padj = 0.097$) were more abundant in the Control group (Figure 4B). The species-level analysis revealed the greatest diversity of signatures, with eighteen differentially abundant viruses, including *Lactobacillus phage phiPYB5* ($\log_2FC = +4.58$; $padj = 0.007$) and Fur seal feces associated circular DNA virus ($\log_2FC = +4.01$; $padj = 0.0009$) enriched in Diarrhea group, and *Porprismacovirus macas1* ($\log_2FC = -3.81$; $padj = 0.002$) and *Phitreivirus phi3* ($\log_2FC = -3.36$; $padj = 0.0002$) predominantly in the Control group (Figure 4C).

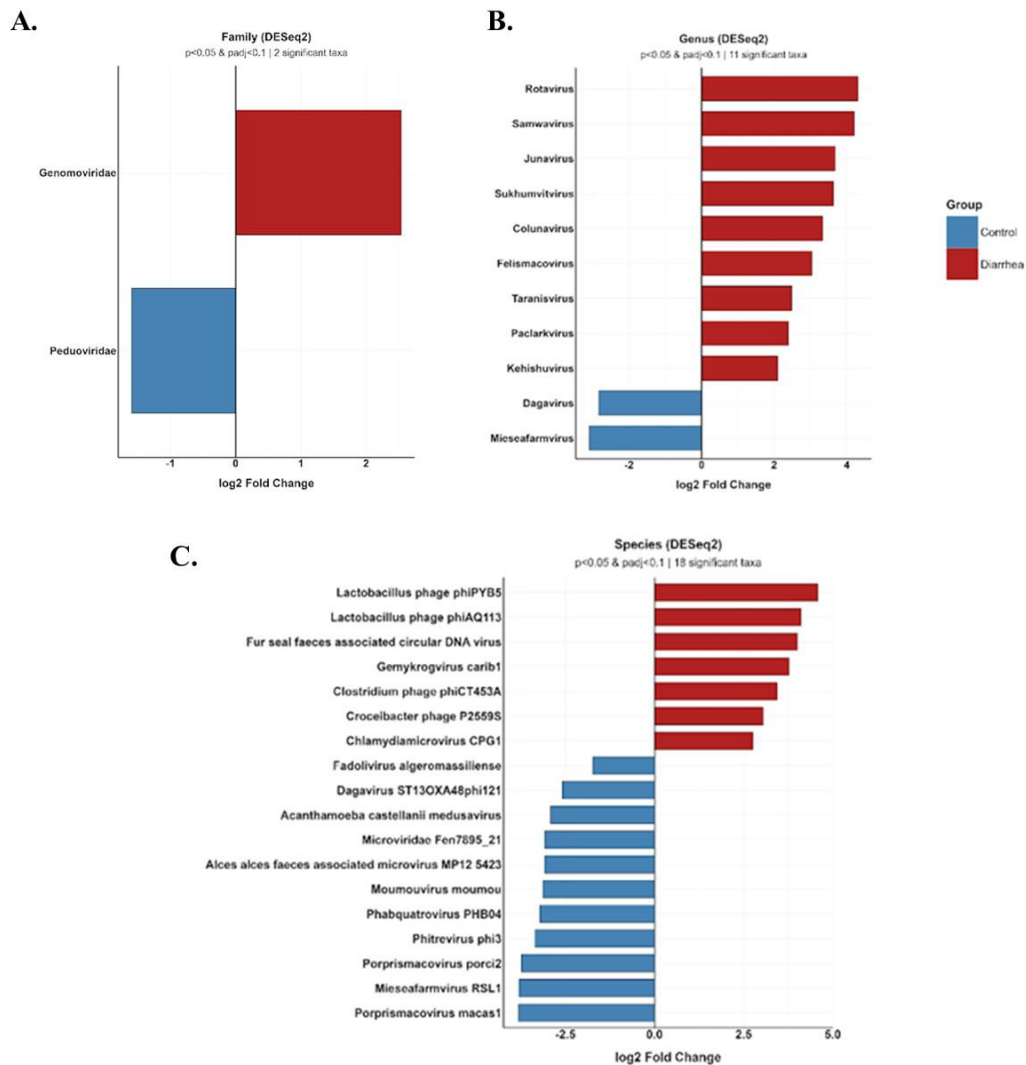


Figure 4. Differential abundance between the Control and Diarrhea groups. The graph shows the viral taxa identified by DESeq2 analysis ($p < 0.05$ and $\text{padj} < 0.1$) with significant differences in abundance. (A) Family, (B) Genus, and (C) Species, respectively.

Discussion

Comparison of the virome between post-weaning diarrhea and the control groups revealed distinct viral profiles. The phylum *Phixviricota*, dominant in relative abundance in the control group, is composed mainly of bacteriophages of the family *Microviridae*, small single-stranded circular DNA (ssDNA) viruses with ubiquitous distribution in aquatic environments and the gastrointestinal tract. Their relevance in intestinal homeostasis is linked to genera such as those of the subfamily *Bullavirinae*, which

includes *Alphatrevirus*, *Gequatrovirus*, and *Sinsheimervirus*, specialized in infecting enterobacteria. By modulating these bacterial communities, these phages contribute to the regulation of the intestinal ecosystem (Kurilovich & Geva-Zatorsky, 2025). This profile was complemented by the detection of viruses belonging to the phyla *Uroviricota* and *Cressdnaviricota*. These groups include circular single-stranded DNA viruses (CRESS-DNA), often found in fecal samples from pigs (Qian et al., 2024). In contrast, the group with diarrhea showed a reduction in the relative abundance of the phyla *Phixviricota* and *Uroviricota* and the family *Microviridae*, suggesting a loss of microbiota modulation by phages (Fehér et al., 2021). At the same time, there was an increase in the phylum *Cossaviricota*, especially in the family *Parvoviridae* and the genus *Bocaparvovirus*, commonly associated with gastroenteritis (Qian et al., 2024).

In the samples from the diarrhea group, the pathogens that stood out with the highest relative abundance at the species level were *Bocaparvovirus ungulate 5* and *Porcine bocavirus 5*. This finding highlights the relevance of these agents in swine enteric conditions and, considering their reported detection in multiple species, corroborates the importance of a One Health approach to understanding their epidemiology, potential for circulation between hosts, and impact on animal production (Prpić et al., 2024; Sarchese et al., 2024). These viruses belong to the family *Parvoviridae*, have a single-stranded DNA (ssDNA) genome and a remarkable range of hosts, including mammals, birds, and insects (Anoyatbekova et al., 2025). While classical *Protoparvovirus ungulate1* remains the leading cause of reproductive disorders (Vargas-Bermudez et al., 2025), our results reinforce the potential role of bocaparvoviruses in the etiology of diarrhea. Porcine bocaparvoviruses, first identified in Sweden in 2009 (Blomström et al., 2009), are now recognized as global pathogens associated with gastrointestinal and respiratory syndromes. These viruses are often described in co-infections with agents such as PCV2, PEDV, and PRRSV (Liu et al., 2024; Tan et al., 2022). The ability of porcine bocaparvovirus to act as a primary pathogen was recently demonstrated in the Harbin (China) outbreak (Ji et al., 2025). Although *Porcine bocavirus 5* has been reported in studies since 2012 (Li et al., 2012), it remains unclassified and is not listed in the ICTV. The detection of *Bocaparvovirus ungulate 5* and *Porcine bocavirus 5* in our study suggests that these viruses may be

relevant to post-weaning diarrhea, although they are probably not the only etiological agent and may act in association with other detected viruses. Thus, they deserve greater attention in the farm environment, and further studies are needed to understand its circulation among animals.

No significant differences in alpha diversity were detected between the control and diarrhea groups, demonstrating the preservation of the overall structure of the viral community. A previous study on the virome of weaned piglets with diarrhea also found no differences in alpha diversity compared to healthy animals (Nantel-Fortier et al., 2022). In contrast, our results of beta diversity analysis show a significant dissimilarity by the Jaccard metric, which revealed a clear distinction between the groups based on the presence or absence of viral taxa. It is important to note that the Bray-Curtis metric, in turn, remained unchanged. These results indicate that diarrhea is mainly associated with changes of specific viral taxa, without compromising the relative proportions of dominant viruses in the community. This suggests that this is a characteristic dynamic of the intestinal virome in diarrheal contexts and relevant for comparative investigations of the virome between sick and healthy individuals (Tao et al., 2022). It is important to note that all animals shared the same zootechnical profile (identical breed, growth stage, and management protocol). This standardization is corroborated by a previous study by our group, which also found no significant differences in intestinal bacterial microbiota diversity (Dias et al., 2025). These data indicate that management can act to standardize microbial communities, so the specific changes observed in the virome are more likely attributable to the clinical condition of diarrhea.

In the diarrhea group, a significant increase in the *Genomoviridae* family was observed, corroborating recent studies that associate these viruses with enteropathies in pigs (Qian et al., 2024; Wang et al., 2025). Their previous detection in cases of human diarrhea and in other species (Zhuang et al., 2023) suggests a potential gastrointestinal tropism or association with dysbiosis through multiple hosts. In contrast, the control group showed significant enrichment of the *Peduviridae* family, tailed bacteriophages such as phage P2, which infect *Enterobacteriaceae* such as *Escherichia coli*, *Shigella* spp., and *Salmonella* spp. (Cunliffe et al., 2022; Fa-arun et al., 2023; Turner et al., 2023). This profile suggests an ecological role for this phage group as a moderator of

enterobacterial populations in the healthy intestine.

At the genus level, *Rotavirus A* stood out as the most distinctive viral signature in the diarrhea group, corroborating its recognized etiological role in piglet diarrhea, although it was also detected in asymptomatic animals (Smol'ak et al., 2022). The circulation of multiple genotypes (A, B, C, H) is widely documented (Krasnikov et al., 2024; Puente et al., 2023), with genotypes A and C predominating in suckling and weaned piglets (Strydom et al., 2024). *Rotaviruses* of genotype A (*RVA*), belonging to the *Reoviridae* family, are characterized by high environmental resistance and tropism for mature enterocytes of the small intestine (Staff, 2025). Infection causes villous atrophy, malabsorption of nutrients, and osmotic diarrhea, resulting in dehydration, weight loss, and, in severe cases, death (Carossino et al., 2024). In pig farming, transmission occurs mainly via the fecal-oral route, with significant productive and economic impact on suckling and weaned piglets. Additionally, pig herds can act as reservoirs, with evidence of zoonotic transmission (Strydom et al., 2024). Control is based on biosecurity, cleaning, and disinfection, although viral stability makes eradication challenging. Vaccination of breeding stock is used to boost humoral immunity and antibody transfer via colostrum, although overly strict health protocols may limit natural exposure and modulate maternal immunity (Rybkowska et al., 2025).

In the diarrhea group, significant enrichment of the genera *Junavirus*, *Sukhumvitvirus*, and *Colunavirus* was also observed, all of which are bacteriophages specialized in *Lactobacillus*. This finding is relevant because, although *Lactobacillus* has probiotic properties, its phages can act as vectors of virulence genes through the process of lysogenic conversion (Lv et al., 2025). By integrating their genome into the bacterial chromosome (forming a prophage), these phages can introduce factors that increase bacterial pathogenicity or convert non-pathogenic strains into pathogenic ones (Pei et al., 2021), which may justify their association with the diarrheal condition.

The genus *Kehishuvirus*, composed of bacteriophages that infect bacteria of the genus *Bacteroides* (fundamental components of the swine microbiota), was also differentially enriched in the diarrhea group. Evidence in humans suggests that the activity of these phages can beneficially modulate the intestinal ecosystem, suppress inflammation and promote epithelial barrier homeostasis (Dinić et al., 2025). Their

enrichment in diarrheic animals may therefore represent a phage response to contain bacterial populations involved in gastrointestinal injury. However further studies are necessary to confirm this potential role.

In contrast, the control group showed enrichment of the genus *Miseaefarmvirus* (species *RSL1*), temperate bacteriophages that infect *Klebsiella pneumoniae*. Their high presence in asymptomatic animals suggests intestinal colonization by *K. pneumoniae* in healthy pigs (Wareth & Neubauer, 2021). This profile could indicate a balanced phage-bacteria interaction in vivo, a relevant hypothesis given the problem of antimicrobial resistance associated with this species (Kot & Witeska, 2024). However, confirmation depends on integrated studies that directly correlate the dynamics of these phages with the abundance and profile of *K. pneumoniae* in the same environment.

Among the species enriched in the control group, *Porprismacovirus macas1* and *Porprismacovirus porci2* stand out. The genus *Porprismacovirus*, initially detected in pigs and primates (Dai et al., 2024), has the widest host range in the *Smacoviridae* family (Fehér et al., 2021). Complementarily, the genus *Felismacovirus* (from the same family) showed enrichment in the diarrhea group. Both are frequently detected in fecal and environmental samples, with no clear association with pathogenicity, and have been described in healthy and sick animals (Liu et al., 2022).

The species *Phitreivirus phi3* was enriched in the control group. This is a bacteriophage specific to *Pseudomonas aeruginosa*, and its enrichment corroborates to previous bacteriological data from these same animals, which reported an increase in the genus *Pseudomonas* in the control group (Dias et al., 2025). This bacteriophage has documented circulation in Brazil (Xavier et al., 2025), and the presence of *P. aeruginosa* in the gastrointestinal tract is feasible due to its ability to spread systemically, although it is commonly more associated with the respiratory tract (Wood et al., 2023). Similarly, the enrichment of the species *Phabquatrovirus PHB04* in the control group, a bacteriophage that infects *Bordetella bronchiseptica*, suggests a modulatory role in the intestinal environment. Although it is a primary respiratory pathogen, *B. bronchiseptica* can cause gastrointestinal lesions in pigs (Yang et al., 2022), and phage therapy has been shown to be effective in controlling it (Hosseindoust et al., 2023).

It is important to point out that this study has limitations inherent to its design

and the methodologies employed. The characterization of the virome itself is limited by the scarcity of reference databases. In addition, the animals that manifested clinical symptoms were immediately separated for treatment. Consequently, it was not possible to correlate the microbial changes observed with the exact duration or severity of individual diarrhea. Thus, the results represent a spot sampling of clinical states within a controlled management context, and the inferences made are, by nature, correlational and indicative.

Conclusion

This study reveals that diarrhea in weaned piglets triggers a qualitative restructuring of the intestinal virome, without a generalized reduction in its diversity. Alpha diversity is maintained, along with a significant change in qualitative beta diversity, indicating that diarrhea does not promote a generalized disruption of the viral community, but rather a selective replacement of certain taxa, suggesting an ecological filter effect that reshapes the viral composition. This reorganization is characterized by the decline of regulatory bacteriophages and commensal viruses, while viruses associated with pathogenicity and dysbiosis become predominant, outlining a distinct viral signature of the diarrheal state. An increase in the relative abundance of *Bocaparvovirus* in the diarrheal group was observed, indicating a need for further studies to its detection in pig farms in evaluated area. We also identified an enrichment of *Rotavirus* in the diarrhea group, an agent widely described as a cause of diarrhea in weaned piglets and known for its high environmental resistance, which makes control in farms difficult. Both viruses have zoonotic potential, and their presence represents an occupational risk for farm workers and reinforces the need to implement more rigorous sanitary measures.

Data availability

The raw sequence data have been deposited in the National Center for Biotechnology Information (NCBI) and are available under BioProject access number PRJNA1358238.

Funding

P.M.R. was a CNPq research fellow (309024/2020-0). F.Q.M. is a CNPq research fellow (305705/2025-3). KKRD has a CAPES doctoral fellowship (88887.664183/2022-00).

Declaration of conflict of interest

None.

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Supplementary Table S1

sample	raw reads	trimmed reads	remaining reads (%)	number of viral reads	viral reads (%)
C1	139278	69349	49.79	4620	6.66
C2	88572	44124	49.81	4511	10.22
C3	144894	72163	49.80	4711	6.52
C4	146010	72731	49.81	3844	5.28
C5	1288676	641894	49.81	67320	10.48
C6	90212	44718	49.56	811	1.81
C7	135248	67466	49.88	6086	09.02
C8	118122	59061	50.00	4752	08.04
C9	113754	56658	49.80	4417	7.79
C10	32420	16070	49.56	4620	28.74
C11	146826	73265	49.89	7100	9.69
C12	119842	59769	49.87	9396	15.72
C13	109678	54682	49.85	17112	31.29
C14	124044	61702	49.74	2197	3.56
C15	123906	61775	49.85	7224	11.69
D1	179032	89290	49.87	2474	2.77
D2	97964	48872	49.88	16063	32.86
D3	104222	51964	49.85	6752	12.99
D4	119162	59386	49.83	8537	14.37
D5	151880	75715	49.85	6956	9.18
D6	143394	71535	49.88	14177	19.81
D7	112016	55847	49.85	18305	32.77
D8	49508	24704	49.89	2580	10.44
D9	90236	44873	49.72	3808	8.48
D10	100542	50171	49.90	9345	18.62
D11	107350	53546	49.87	2474	4.62
D12	96938	48360	49.88	12866	26.60
D13	130340	65000	49.86	6306	9.70
D14	126308	62891	49.79	3917	6.22
D15	86416	43037	49.80	3372	7.83

6. DISCUSSÃO GERAL

A diarreia pós-desmame na suinocultura é um distúrbio de etiologia multifatorial, envolvendo complexas interações entre o hospedeiro, ambiente e o microbioma intestinal. Para elucidar os fatores microbianos envolvidos na patogênese, este estudo analisou a microbiota bacteriana e o viroma de leitões diarreicos e saudáveis. Apesar da diversidade global da microbiota se manter estável, identificamos alterações pontuais e significativas na composição de ambos os componentes nos animais doentes. Tais descobertas fornecem evidências que podem auxiliar na compreensão sobre as disbioses bacterianas e virais que possivelmente tenham relação com o surgimento da diarreia em leitões em fase de desmame.

Nos animais com diarreia, observou-se um aumento na razão *Bacillota/Bacteroidota*. Essa razão tem sido amplamente utilizada como biomarcador de saúde intestinal (Karačić et al., 2024), embora sua interpretação seja complexa, uma vez que estudos há estudos discordantes relacionados a diarreia e outras patologias (Zhou et al., 2025; Hu et al., 2023), assim como a redução dessa razão também já foi caracterizada como benéfica para microbiota intestinal (Goggans et al., 2022).

Não foram detectadas alterações significativas na diversidade alfa da microbiota bacteriana e do viroma intestinal, demonstrando a estabilidade da diversidade intrínseca dessas comunidades microbianas, mesmo na presença de sinais clínicos. Em contraste, a análise de diversidade beta revelou um perfil distinto: enquanto a composição bacteriana permaneceu similar entre os grupos, o viroma exibiu uma separação significativa no índice de Jaccard. Esse resultado aponta para uma dissimilaridade baseada na composição qualitativa (presença/ausência) de vírus específicos nos animais diarreicos, mas sem diferença nas proporções quantitativas virais.

Na abundância diferencial o grupo controle apresentou um perfil composto por bactérias com propriedades probióticas (*Liquorilactobacillus*) (Liu et al., 2023) e bactérias fermentadoras (*Komagataeibacter*, *Acetobacter* e *Gluconobacter*), que auxiliam na digestão dos alimentos e atuam na modulação intestinal (Choi et al., 2022). Também foram observadas bactérias oportunistas enriquecidas (*Escherichia/Shigella*, *Salmonella*, *Pseudomonas*), indicando uma possível colonização controlada (Wang et al., 2021). Já

em relação os vírus enriquecidos, foi observado no grupo controle bacteriófagos de enterobactérias (*Peduviridae*) e outros bacteriófagos de bactérias respiratórias que já foram descritos presentes no trato gastrointestinal (*Phitrevirus phi3* e *Phabquatrovirus PHB04*), a presença desses bacteriófagos sugere um papel ativo no controle de bactérias oportunista e equilíbrio da comunidade microbiana, auxiliando na homeostase intestinal (Cunliffe et al., 2022; Wood et al., 2023; Yang et al., 2022) .

Em contraste, no grupo diarreia estavam enriquecidas bactérias com potencial inflamatório (*Fusobacterium*) e outra bactéria respiratória já descrita em leitões diarreicos (*Pasteurella*), mostrando a presença de bactérias oportunistas, que podem intensificar quadros de diarreia em suínos (Kong et al., 2022; Luo et al., 2022) . Em relação aos vírus mais abundantes, destacou-se o enriquecimento *Rotavirus A*, com etiologia já bem descrita na literatura em leitões diarreicos, que além de possuir grande resistência ambiental também tem potencial zoonótico (Staff, 2025; Strydom et al., 2024).

Além disso, o aumento da abundância relativa no grupo diarreia de *Bocaparvovirus ungulate 5* e *Porcine bocavirus 5*, agentes emergentes presentes em casos de diarreia em diversas espécies de hospedeiros, incluindo humanos, mostra a necessidade de uma abordagem *One Health* para compreender melhor a patogenia desse grupo viral (Prpić et al., 2024). *Bocaparvovirus* já foram descritos em muitos trabalhos atuando em conjunto com outros vírus em casos de diarreia (Liu et al., 2024), porém um estudo recente chinês mostrou a presença desse vírus como único agente em um surto de diarreia em suínos (Ji et al., 2025). Essas evidências destacam a necessidade de se implementarem técnicas diagnósticas mais específicas para esse agente e de mais estudos para compreender a circulação, patogenicidade e o papel clínico preciso desses vírus.

Esta tese apresenta limitações em duas frentes principais. Na caracterização clínica, não foi possível obter registros detalhados sobre a evolução dos sintomas, especialmente o tempo de duração da diarreia, o que dificulta a correlação entre o quadro clínico e os agentes infecciosos identificados. Nas análises de bioinformática, não foi realizada a montagem dos genomas virais a partir dos dados de sequenciamento e uma análise conjunta da microbiota bacteriana e viral. Essas lacunas restringem uma compreensão mais completa das interações entre os patógenos e abrem caminho para investigações futuras que combinem dados clínicos refinados com abordagens integradas

de metagenômica.

7. CONCLUSÕES

Este trabalho identificou diferenças pontuais, porém significativas, na composição bacteriana e viral entre leitões saudáveis e com diarreia pós-desmame, apontando para possíveis causas microbianas específicas associadas ao quadro clínico. Foi observada uma distinção na composição dos perfis microbianos, com o grupo saudável apresentando uma comunidade bacteriana e viral potencialmente protetora e o grupo doente exibindo o enriquecimento de bactérias oportunistas e vírus entéricos, reforçando a complexa interação entre esses componentes na saúde e na doença em leitões em fase de desmame.

Não foram encontradas diferenças na diversidade alfa (microbiota e viroma) nem na composição bacteriana (beta). Porém, a presença/ausência de vírus diferiu significativamente entre os grupos, indicando que a diarreia pode estar associada a uma alteração específica no perfil viral, sem mudanças expressivas nas comunidades bacterianas.

O perfil metabólico funcional bacteriano também divergiu entre os grupos. No grupo saudável, as vias predominantes sustentam funções fisiológicas e de crescimento. Já nos leitões diarreicos, observou-se o enriquecimento de vias metabólicas bacterianas específicas, indicando uma maior atividade bacteriana que pode estar intensificando a patogênese da diarreia.

A detecção de agentes com reconhecido potencial zoonótico, como *Rotavirus A* e *Bocaparvovirus*, somada à presença de patógenos zoonóticos oportunistas, como *Fusobacterium*, transcendem o âmbito da saúde animal e reforça a necessidade de uma abordagem *One Health* para compreender a patogênese, a circulação e o impacto epidemiológico desses patógenos. A identificação de tais microrganismos em suínos destaca a interface entre a saúde animal e a saúde pública. Portanto, estratégias futuras de vigilância, prevenção e controle devem ser estudadas, reconhecendo que a elucidação das interações dinâmicas entre microbiota e viroma é fundamental tanto para a sanidade suinícola quanto para a saúde global.

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CURRICULUM VITAE RESUMIDO

DIAS, K.K.R
DIAS, KAREN KARINE DA ROSA

1. DADOS PESSOAIS

Nome: Karen Karine da Roda Dias

Local e data de Nascimento: Porto Alegre, Rio Grande do Sul, Brasil, 26 de outubro de 1989.

Endereço profissional: Centro de Biotecnologia, Avenida Bento Gonçalves, 9500, Setor 4, Prédio 43421, laboratório 206, Porto Alegre, RS; CEP 91509-900.

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2. FORMAÇÃO

Curso de graduação: Medicina Veterinária, Universidade Federal do Rio Grande do Sul (UFRGS). Período: 2014-2018.

Curso de graduação: Medicina Veterinária, Universidade Luterana do Brasil (ULBRA). Período: 2011-2013.

Curso de Pós-Graduação concluído: Mestrado *stricto sensu* pelo Programa De Pós-Graduação em Biologia Celular e Molecular Aplicada à Saúde (PPGBioSaúde), ULBRA. Período: 2019-2021.

Título: Análise Filogenética e Perfil de Genes de Resistência a Antimicrobianos de Isolados de *Salmonella enterica* do Sorotipo *Typhimurium* de Animais de Produção e Alimentos no Sul do Brasil

Orientador: Prof. Dr. Vagner Ricardo Lunge

Curso de Pós-graduação em andamento: Doutorado *stricto sensu* pelo Programa De

Pós-Graduação em Biologia Celular e Molecular (PPGBCM), UFRGS. Período: 2022-2026.

Título: Caracterização da Microbiota Bacteriana e do Viroma Intestinal de Leitões com e Sem Diarreia Pós-Desmame.

Orientadora: Prof^ª Dr^ª Fabiana Quoos Mayer

2.1 FORMAÇÃO COMPLEMENTAR

17º Simpósio Internacional de Suinocultura - SINSUI 2025. Carga horária: 22h.

Período: 2025-2025. Centro de Eventos PUCRS, Porto Alegre, Brasil.

Curso Inglês. Carga horária: 73h. Período: 2024-até o momento. Fernanda Macário-Inglês Personalizado, Porto Alegre, Brasil.

Curso de Bioinformática Aplicada à Metagenômica: Análise Integrativa de Dados Microbianos. Carga horária: 45h. Período: 2024-2024. Online. Rede de Pesquisa dos ecossistemas Mangue e Restinga (Rede MaRe), Brasil.

VI Curso de Verão em Biologia Evolutiva na UFRGS. Carga horária: 30h. Período: 2020-2020. Universidade Federal do Rio Grande do Sul, Porto Alegre, Brasil.

3. ESTÁGIOS

Vínculo: Estágio Curricular em Clínica de Pequenos Animais Domésticos. Poa Pet Care Centro Veterinário. Porto Alegre, Brasil. Período: 2018-2018.

Orientadora: Cristiane dos Reis Ritter

Vínculo: Estágio Curricular em Fisioterapia Veterinária. Mundo à Parte de fisioterapia. Porto Alegre, Brasil. Período: 2018-2018.

Orientadora: Denise Silva Provenzano Lima

Vínculo: Monitoria Acadêmica Disciplina de Medicina de Suínos. Setor de Suínos-SETSUI. Instituição: UFRGS. Porto Alegre, Brasil. Período: 2017-2018

Orientador: Prof Dr Rafael da Rosa Ulguim

Vínculo: Estagiária plantonista. Terra dos Bichos. Porto Alegre, Brasil. Período: 2016-2017.

Vínculo: Bolsista de Extensão. Setor de Toxicologia Veterinária. Departamento de Patologia Clínica Veterinária. . Porto Alegre, Brasil. Período: 2016-2016.

Orientador: Marcelo Dutra Arbo

Vínculo: Estagiária plantonista. Poa Pet Care Centro Veterinário. Porto Alegre, Brasil. Período: 2015-2016.

Vínculo: Bolsista de Iniciação Científica. Laboratório de Embriologia e Biotécnicas de Reprodução Instituição: Universidade Federal do Rio Grande do Sul, UFRGS, Brasil. Período: 2014-2015.

Projeto: Avaliação da expressão gênica em embriões bovinos produzidos *in vivo* e em diferentes sistemas de cultivo *in vitro*.

Orientador: Prof Dr Jose Luiz Rigo Rodrigues

Vínculo: Estagiária plantonista. Hospital Veterinário Lorenzoni. Porto Alegre, Brasil. Período: 2014-2014.

Vínculo: Estagiária de análise clínicas. BLUTS Centro de Diagnósticos Veterinários. Porto Alegre, Brasil. Período: 2014-2014.

Vínculo: Estagiária plantonista. Poa Pet Care Centro Veterinário. Porto Alegre, Brasil. Período: 2013-2014.

Vínculo: Monitoria Acadêmica Disciplina Anatomia Veterinária 2. Setor de Anatomia Veterinária. Instituição: ULBRA. Canoas, Brasil. Período: 2017-2018

Vínculo: Bolsista de Iniciação Científica. Laboratório de Diagnóstico Molecular

Instituição: ULBRA Canoas. Brasil. Período: 2013-2013

Projeto: Estudo da disseminação de genótipos do vírus da bronquite infecciosa das galinhas em lotes de produção industrial de aves.

Orientador: Prof Dr Nilo Ikuta

Vínculo: Bolsista de Iniciação Científica. Laboratório de Diagnóstico Molecular

Instituição: ULBRA Canoas. Brasil. Período: 2012-2013

Projeto: Estudo da diversidade genética de mycoplasma gallisepticum e mycoplasma synoviae em lotes de produção industrial da região sul e sudeste do brasil.

Orientador: Prof Dr Edmundo Kanan Marques

Vínculo: Estagiária de diagnóstico por imagem. Divet. Porto Alegre, Brasil. Período: 2012-2013.

4. PRÊMIOS E DISTINÇÕES

Menção Honrosa do Prêmio de Pesquisa Avícola "José Maria Lamas da Silva" para o trabalho "*Salmonella Gallinarum*: Diferenciação Molecular da Cepa Vacinal SG9R de Isolados de Campo", Fundação APINCO de Ciência e Tecnologia Avícolas. 2014

Destaque no XXV Salão de Iniciação Científica da UFRGS com indicação ao prêmio Jovem Pesquisador, Universidade Federal do Rio Grande do Sul. 2013

5. ARTIGOS COMPLETO PUBLICADOS

DIAS, KAREN KARINE DA ROSA; TOCHETTO, CAROLINE; VARELA, ANA PAULA MUTERLE; LOIKO, MÁRCIA REGINA; DOS SANTOS, ANNE CAROLINE; FRAZZON, JEVERSON; ROEHE, PAULO MICHEL; MAYER, FABIANA QUOOS. Impact of weaning on piglet microbiota: differences in bacterial taxa

and metabolic pathways associated with diarrhea. MICROBIAL PATHOGENESIS, v. 207, p. 107878, 2025. DOI: 10.1016/j.micpath.2025.107878

GUEDES, M. T. J. C.; RIBEIRO, M. H. G.; SOROKA, V. D.; **DIAS, K. K. R.**; FRAZZON, J.; SILVA, M. C. A. Assessment of Methane Production and Microbial Community Dynamics in One and Two-Stage Semi-continuous Anaerobic Systems Treating Orange Peel Waste and Crude Glycerol. BioEnergy Research, v. 19, p. 48, 2026. DOI: <https://doi.org/10.1007/s12155-026-10977-6>

SILVA, C. B.; ZAMPAR, A.; DANIELI, B.; NASCIMENTO, A. L.; BAVARESCO, L. H.; RIGO, E.; KLEIN, B.; TRIBST, A. A. L.; **DIAS, K. K. R.**; MAYER, F. Q.; VARELA, A. P. M.; MANN, M.; FRAZZON, J.; ENDRES, C. M.; SCHOGOR, A. L. B. Effects of Incorporation of Essential Oils in the Jersey Cow Diet on the Quality of Produced Dairy Products (Milk, Cream, and Colonial Cheese). Foods, v. 14, p. 2788, 2025. DOI: <https://doi.org/10.3390/foods14162788>

ZANETTI, N S; DE CARLI, S; SOUZA, M N; LEHMANN, F K M; KIPPER, D; **DIAS, K K R**; FONSECA, A S K; LUNGE, V R; IKUTA, N. Molecular Detection and Characterization of Salmonella Gallinarum From Poultry Farms in Brazil. JOURNAL OF APPLIED POULTRY RESEARCH, v. 1, p. 1-7, 2019. DOI: <https://doi.org/10.3382/japr/pfz051>

6. RESUMOS E TRABALHOS APRESENTADOS EM CONGRESSOS

DIAS, K K R; TOCHETTO, C.; VARELA, A. P. M.; LOIKO, M. R.; SANTOS, A. C.; FRAZZON, J.; ROEHE, P. M.; MAYER, F. Q. Diarreia no desmame e sua relação com a microbiota intestinal de leitões: uma análise comparativa. In: Anais SINSUI XVII, 2025, Porto Alegre. Anais do SINSUI 2025 ? 17º Simpósio Internacional de Suinocultura.

Porto Alegre - RS - Brasil, 2025. v. 17.

MACIEL-FIUZA, M. F.; SBRUZZI, R.C.; ROSSET, C.; EIDT, L.M.; MORAES, P.C.; CATTANI, C.A.S.; MAYER, F.Q.; **DIAS, K. K. R.**; VIANNA, F.S.L. Caracterização da microbiota intestinal e correlação com expressão de citocinas pró-inflamatórias em pacientes com hanseníase. In: 17º Congresso Brasileiro de Hansenologia, 2023, Cuiabá. Caracterização da microbiota intestinal e correlação com expressão de citocinas pró-inflamatórias em pacientes com hanseníase, 2023.